

**WOMEN IN COMPUTING CAREERS--DISCOVERING THE FACTORS THAT
ATTRACT AND RETAIN THEM: A QUALITATIVE RESEARCH STUDY**

**A Dissertation Presented in Partial Fulfillment of the
Requirements for the Degree of
Doctor of Management**

By

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Abstract

Despite the increasing number of women entering into science, technology, engineering, and math occupations, the number of women entering computing occupations has drastically fallen since the early 1990s (National Center for Women in Technology, 2014). The purpose of this study was to explore strategies that technology managers can use to retain women in the computing industry. The study conducted was a qualitative exploratory study that was designed to develop strategies that technology managers could use to retain female computing professionals and realized the benefits of a gender diverse team. Data collection occurred using semi-structured interviews, which were analyzed and synthesized to derive conclusions. The findings of this study largely corroborated key elements of the existing literature and led to the identification of eleven strategies for use by technology managers to help improve the engagement and retention levels of female computing professionals.

Keywords: women in computing, women in technology, gender balance in computing, diversity in computing

Dedication

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For Isla, who at four years old can use an iPad better than her father can after an almost twenty year career in computing. I hope that not only can you achieve success in whatever career you choose, but that you can be recognized for the value that you bring to the team just by being uniquely you. Most importantly, for Kelly. You've supported me in ways that nobody else would or could. You've pushed me when I needed pushing, refocused me when I was distracted, and made me feel like I was doing something worthwhile. I can't begin to thank you for everything that you have done and continue to do for me. You will forever be my person.

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During my doctoral journey, I have had the honor to interact with many outstanding people who played key roles in the completion of my doctoral research. First, to the women who agreed to participate in my study, there is a famous saying “Be the change that you want to see in the world.” Whether you set out to make a change, follow your dream job, or just make a dollar or two, you are the change the industry needs. I can’t thank you enough.

I would also like to acknowledge the members of my doctoral cohort. Each has been a source of inspiration, insight, and support. Crossing each other’s path in-person (at the symposia) or online (during our live chats) have added to and enhanced my doctoral journey. They are an exceptional group of people who are about to do exceptional things as they continue and complete their doctoral journeys. Specifically Dr. Larry Solliday, Dr. Stefany Grieco, and Dr. Rodney Montoya. You all helped me get across the finish line. Especially you Larry, you were my compass when I was feeling incredibly lost.

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CHAPTER ONE

Developing strategies to retain women in the computing industry have become increasingly important for technology managers due the continued decline of women in the industry (National Center for Women in Technology, 2014), especially in the face of a shortage of skilled professionals in key computing fields, such as Big Data and Cybersecurity (Manyika et al., 2011; Kauflin, 2017). Computing jobs also represent a higher paying jobs, with women earning more than women in other professions (Funk & Parker, 2018a), which would go far in addressing the 25% pay gap between genders identified by the American Association of University Women (2014). Thus, technology managers have many important responsibilities, including the development of strategies that are successful in attracting and retaining female computing talent (Beyer, 2014).

The goal of this study is to understand, from the standpoint of women within the computing industry, what strategies are most helpful in overcoming deterrents or barriers to entry for women seeking careers. The significant meaning of collecting data directly from women who have overcome the identified barriers is critical in motivating future generations of women facing those same barriers (Walters, 2015). Bowles (2012) pointed out that scholars of gender have no problem identifying the struggles that women in industry face, however she argued that those same scholars often struggle to identify those reasons that allow women to succeed. By directly collect collecting success stories from women in computing that have achieved those successes it may be possible to transform them into viable strategies that technology managers can employ to replicate those successes.

Organizations, both academic and professional often struggle to maintain women, with as many as 45% of women leaving the computing industry within their first five years (Barr, 2017).

The struggle is often due to the common issues, such as self-efficacy (Beyer, 2017; Govender & Khumalo, 2014; Linn, 2016) or identity alignment (Lewis, Anderson, & Yasuhara, 2016). Thus, understanding how women have successfully overcome these issues would be instrumental to both attracting and retaining female talent (Walters, 2015).

It is evident from the existing literature that there are fewer women than men working in the computing industry. This literature also pointed out that the number of women in other Science, Technology, Engineering, and Math (STEM) disciplines are not only higher, but have been steadily increasing, while the number of women in computing has declined. With the U.S. Census Bureau (2017) reporting that almost half of the American workforce comprised of women, it begs the question of why that same balance of genders does not occur in the computing industry.

This chapter provides an overview of the topic of gender diversity in computing, as well as identify the problem and purpose statements. Additionally, the chapter examines the research question, with a discussion of the theoretical frameworks that will guide the research process. Lastly, the chapter also discusses study assumptions, boundaries, and limitations, and definitions of specific terminology.

Topic Overview/Background

A stark contrast exists between the number of women in computing and women in other STEM careers, according to the National Center for Women in Technology (NCWIT; 2014). The contrast is visible in the statistics. For example, in 2013, women comprised 59% of the workers in the American biological science industry (NCWIT, 2014). The NCWIT also reported that women also accounted for 39% of American chemists and material scientists during that same period. According to the Bureau of Labor Statistics (2017) only 25% of American computing professionals were women.

One major contributor to the gender gap in the computing industry is that fewer women pursued degrees in computing than men. Schimpf, Andronicos, and Joyce (2015) argued that the fact that women only accounted for 30% of computer science degrees was indicative of the much larger, systemic problem of how women interact with computers in their formative years. Coupled with the additional issues of self-alignment with perceived stereotype (Alvarado & Judson, 2014; Beyer, 2014; Bock, Taylor, Phillips, & Sun, 2013; Cundiff & Vescio, 2016; Lewis, Anderson, & Yasuhara, 2016) and inconsistent institutional support (Cohoon, 2011), it was clear that there were several issues that acted as barriers to entry from women seeking employment in the computing industry.

Measures to foster increased involvement from women in the industry must not do so at the exclusion of men; any solution requires participation from both female as well as male genders. Calls for male advocates from the computing industry have been made (Farmer, 2014; Herscher, 2014) and answered. In a panel at the 2014 Grace Hopper Celebration, Eustace from Google, Stansbury from Intuit, Schroepfer from Facebook, and Irving, from GoDaddy all spoke on initiatives to help improve the numbers of women in technology (Herscher, 2014).

Despite the fact there have been numerous studies conducted regarding the gender balance in computing, Vitores and Gil-Juarez (2016) argued that very little has changed with the situation, with the number of women seeking careers in the computing industry continues its decline.

Problem Statement

The problem to be addressed in the proposed study is the strategies technology managers need to retain women in the computing industry have not been identified (Beyer, 2014). According to the US Department of Labor Bureau of Labor Statistics (2017), in 2017 the number total number of working Americans was 157 million with women comprising approximately

47% of the workforce. Even though women constitute almost half of the overall workforce, a 2013 study showed that only 26% of computing industry workers are women (National Center for Women in Technology, 2014). In 2017, women comprised only 25.5% of the 4.8 million computing professionals (Bureau of Labor Statistics, 2017), which indicates that the number is continuing its decline. The disparity in gender is not only alarming, but it also begs the question of what factors contribute to the skewed numbers and whether or not those factors could equally affect other industries.

In addition to the shortage of female IT candidates, technology managers face issues with retaining the female employees that they do attract. Cundiff, Vescio, Loken, and Lo (2016) pointed out that the problem of self-alignment with perceived stereotype persists after college. Women also face an unconscious bias in the workplace that places their technical proficiency at a lower level than their male counterparts (Anita Borg Institute, 2013). Retention is also a key concern of technology managers. Research shows that replacing an employee could cost as much as 60% of their annual salary. The intangible costs of turnover, such as knowledge loss, innovation, and competitiveness should also be of more significant concern (Anita Borg Institute, 2013).

Lastly, technology managers are facing a shrinking pool of female candidates to help them create a diverse gender team. According to Corbett in Hill (2015), women comprised on 19% of computer science graduates in the United States. Even those technology managers that are proactively seeking qualified female candidates may have difficulty finding, hiring, and retaining them.

Purpose Statement

The purpose of the proposed qualitative exploratory study is to explore the strategies technology managers need to retain women in the computing industry. There are estimated

shortages in crucial computing professions, such as cybersecurity, which Kauflin (2017) estimated as having a shortage of two million qualified candidates by 2019. Another growing computing discipline, big data, could face shortages of 140,000 – 190,000 workers with the appropriate skillset, according to Manyika et al. (2011). These shortages, amongst others in the industry, represent well-paying jobs that could potentially be filled by women.

Research Question

With the problem and purpose of the research identified, it is possible to create a central research question, at a broad level, to help guide the research (Creswell, 2014). According to Creswell (2014), a central research question helps researchers explore the intricate factors that surround a phenomenon. The central research question in this study is:

What are the strategies technology managers need to retain women in the computing industry?

Proposition

The research study will identify those positive experiences of women who have successfully obtained employment in the computing industry. Based on the findings of other studies (Bowles, 2012; Walters, 2015), first-hand experience of successful women in an industry that has shown to be difficult for women to succeed in, goes a long way towards helping other women to overcome common deterrents or barriers to entry for that industry. Careers in computing have proven to be a challenge for many women (NCWIT, 2014).

Between 1970 and 1990, women working in the computing industry rose at rates comparable to women working in other comparable industries, such as scientists and engineers (Corbett & Hill, 2015). From 1990 onward, Corbett and Hill point out that the number of women dropped significantly, while the number of women working in comparable industries continued to increase at a steady rate or, at a minimum, remained stable. The decline of women in

computing represents multiple problems for technology managers. In addition to the aforementioned lack of skilled computing professionals, those managers are also unable to take advantage of several benefits that are afforded to teams that have gender diversity (Derven, 2014; DuBow, 2013; Gratton, Kelan, Voight, Walker, & Wolfram, 2007).

Conceptual Framework

The research conducted in this study uses a four-layer conceptual framework that highlights categorical barriers that prevent women interested in computing from attaining both higher education and a career in computing and is based upon the Expanded Social Cognitive Career Model for Attrition/Persistence (Singh & Fouad, 2011). The social cognitive career model (SCCT), according to Singh and Fouad, can be used to predict how relationships between self-efficacy, expectations, and goals and uses that relationship can affect career development. The expanded SCCT model presented by Lent et al. (2003) introduced the elements of barriers and supports, which directly links barriers to career choice through self-efficacy. The four layers in the framework are women interested in the computing industry, educational barriers that prevent women from attaining computing degrees, workplace barriers that prevent women from obtaining and remaining in jobs in the computing industry, and women that succeed at finding and maintaining employment and in the computing industry.

The study focuses on a gap in the literature regarding positive influences on as well as how women socially identify with jobs in the computing industry. This topic lends itself to the social construction of technology theory developed by Pinch and Bijker (1987), which argued that technology is generated based on social influence, which, according to Wajcman (2000), was done with very little input from women. Additionally, the topic aligns with Tajfel and Turner's (1979) social identification theory, which asserts that membership of social group influences individual choice and behavior. Specifically, the way that positive experience may

mitigate previously identified barriers to entry, such as self-identity alignment and competition for access to resources (Alvarado & Judson, 2014; Beyer, 2014; Bock, Taylor, Phillips, & Sun, 2013; Cundiff & Vescio, 2016; Lewis, Anderson, & Yasuhara, 2016) and lack of role models (Gamm, 2014) in the educational realm. Additionally, workplace barriers which include many of the same educational barriers, but also introduce implicit bias barriers (Klein & Díaz-Hernández, 2014).

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Assumptions/Biases

The study has some assumptions and biases:

Statements that are presumed to be true for a specific purpose are considered assumptions (Wargo, 2015). The most significant assumption made in research on the decline of women in the computing industry is that women are interested in pursuing careers in the industry. The premise refers to the fact that before 1990, the number of women seeking careers in computing was increasing at the same rate as other STEM disciplines. Of equal relevance, while the number of women seeking computing careers has dropped, the number of women seeking careers in other STEM disciplines has continued to increase at a steady rate (National Center for Women in Technology, 2014). While the data suggests that interest remains, it is possible that computing jobs are not as desirable as other alternatives.

Simundic (2013) defined bias in research as any trend or deviation from the truth in data collection, data analysis, interpretation and publication, which can cause false conclusions. The study accounts for the existence of the following bias on the part of the researcher. The most prevalent bias in the research is a gender difference between the researcher and the subjects. Being male researching factors that affect females represents a significant gap in experience. However, the literature suggested that to help solve the problem presented, male advocacy in the

industry will go a long way in supporting the cause (Ashcraft, DuBow, Eger, Blithe, & Sevier, 2013). This type of bias is implicit bias.

The second bias in the research relates to the fact that this researcher has had an extensive career in the computing industry, both as an employee and as a manager, that has for the most part been supportive. So far, there have not been experiences that call into question whether remaining in the industry is the best choice. Therein lies the potential for failure to recognize a key factor affecting the research audience due to acceptance of what is perceived to be an industry norm. This type of bias is confirmation bias.

The third bias present is the belief that gender biases are the cause of the gender imbalance in the computing industry. Research has generated evidence that fewer women are seeking degrees in computer science (Schimpf et al., 2015) and that the number of women with careers in the computing industry has fallen by approximately 10% since 1990 (National Center for Women in Technology, 2014). The literature on the topic has proven many factors that contribute to these numbers, but what has yet to be conclusively proven is that gender bias, whether explicit or implicit, adds to the gender balance.

Significance of the Study

The lack of women in the computer industry is worth studying for several reasons. First, it has been well established that diversity, regardless of industry, provides significant advantages, such as improved success from a gender-balanced executive board (Hersh, 2016) and increased engagement and retention (Derven, 2014). Additionally, DuBow (2013) argued that diversity regarding gender, race, and culture are related to improvements in problem solving and creativity. More specifically, a London Business school study that indicated that the closer a team was to having gender balance, the higher the teams performed in the area of experimentation and efficiency (Gratton et al., 2007).

Second, the answers to the research question can help managers fulfill their duties to support the minority members of their subordinates. Karpowitz, Mendleberg, and Shaker (2012) argued that according to their gender role hypothesis that women have lowered status and are less participatory when they represent the minority population of a group. The greater management community should be mindful of this phenomenon because according to Taylor (1914) of equal importance to employer success, is employee success. Additionally, according to Taylor's (1916) views regarding maximum prosperity, it is apparent that management, on the whole, may be failing to secure said prosperity for female employees.

Delimitations

Simon (2011) describes delimitations as controllable factors that can be used to set the scope and boundaries of research. The proposed study has two delimitations. Delimitation 1 is that the study will focus on women who work in the computing industry. Delimitation 2 is that the study will only include women who reside in San Diego, CA.

Limitations

According to Simon (2011), limitations are weaknesses of a study, often uncontrollable factors that have the potential to influence results. The study has several limitations. The first limitation is the willingness of the participants to answer questions honestly. The second limitation of the study is that the participants will feel comfortable in sharing their experiences with a member of the opposite gender.

Definition of Terms

Cognitive bias: A cognitive bias is a mistake in reasoning in which one's preferences or beliefs alter their memory or judgment regardless of contrary information (Cherry, 2018).

Confirmation bias: A confirmation bias is a human tendency to collect, interpret, or otherwise alter information to confirm a pre-existing belief (Heshmat, 2015).

Gender bias: Gender bias is the prejudice or preference for one gender over another (Hill & Hill, 2005).

Implicit bias: An implicit bias is the unconscious alteration of thoughts and decisions based on perceived stereotypes (Payne, Niemi, & Doris, 2018).

Information Technology (IT): IT is the study or use of systems (especially computers and telecommunications) for storing, retrieving, and sending information (Rouse, 2015).

Science, Technology, Engineering, and Mathematics (STEM): STEM is the acronym used commonly to refer to the named fields, which refer to related professional industries (Rouse, 2013).

Social Cognitive Career Model (SCCT): predicts the relationships among self-efficacy, outcome expectations, interests, choice goals and choice actions as they might affect the career development process (Lent et al. 2003).

General Overview of the Research Design

The overall goal of this study was to understand the experiences of women in computing to understand what similarities and differences they face while pursuing a career in computing. The qualitative exploratory research methodology was selected to help understand this because, according to Mansourian (2008), this method is especially useful when dealing with ambiguity in research or when there are uncertain elements within the topic being research. While there is no shortage of theories or hypothesis as to what deters women from seeking education and employment in the computing industry, the continued decline represented in the data suggests that a means to correct this decline has not been found or implemented by current technology managers.

The data collection technique selected to answer the research question will be semi-structured interviews. Audio from the interviews will be recorded and transcribed for use in the analysis phase, along with notes and observations from the interviewer.

Data analysis will consist of activities in three general areas: coding, categorizing and memoing. The coding process follows an epistemological process in which phenomenon are identified and categorized for use in drawing conclusions and developing hypotheses. Categorization activities include grouping codes based on emergent themes and specific concepts. During the interview process, notes and observations of the process will be recorded by this researcher using memos. During the data analysis process, these notes and observations will be revisited to provide contextual information for use in the analysis.

Summary of Chapter One

The purpose of this chapter was to provide a basic understanding of the issue of the lack of a gender-balanced computing industry, as well as to highlight why a study of this topic would be beneficial. A brief discussion of the research methodology was presented along with the terms and definitions that will be used throughout this dissertation.

The next chapter will be a review of the literature that is pertinent to the topic of women in computing, which allows for the creation of a conceptual framework that will be used to understand the complex relationships that associate with the identified phenomenon.

Organization of Dissertation

Chapter 1 posed an overview of the identified problem, the purpose of the study, the research methodology, specific definitions or terminology used, and a breakdown of the five chapters. Chapter 2 is an extensive review of the pertinent literature that discusses the topic of gender diversity in computing. This chapter also contains a conceptual framework for understanding the identified problem. Chapter 3 covers the research tradition and methodology

used, with rationale, the setting, and context of the research, target population and sample selection, data collection and analysis methods, establish trustworthiness and addresses human subjects protection concerns. Chapter 4 summarizes the collected data and the qualitative analysis of the strategies technology managers need to retain women in the computing industry. Chapter 5 summarized Chapter 4 results and presented the conclusions and recommendations for future research.

CHAPTER TWO

The last chapter provided an overview of the topic of this dissertation, namely that there is a documented gender gap in the computing industry. Of additional noteworthiness, there is concern about an upcoming shortage of qualified computing professionals to fill positions in the ever-growing computer industry in critical specialties, such as data science (Manyika et al., 2011). One goal of this study is to redress this imbalance by investigating the concept of social identity or how women perceive themselves as other members of the computing profession. Additionally, it takes a critical look at the experiences of women seeking a degree and employment in the computing industry through a cultural lens.

Chapter 2 reviews the academic literature regarding the subject of gender in the computing industry, with particular emphasis on the existing theories of why women struggle to achieve educational and professional goals in computing. The chapter begins with the discussion of several theories that relate to the evolution of technology, women's role in that evolution, and the importance of social identity towards educational and career pursuits. Next discussed is the importance of gender diversity and how the lack of diversity in computing may exacerbate the issue. Lastly, discussion occurs on specific barriers to entry that women face in both computing educational and employment pursuits.

Social Shaping of Technology and Social Identity

MacKenzie and Wajcman (1985) developed the concept of social shaping of technology (SST) to illustrate the means by which technological innovations are guided. They argued that there were a variety of factors that influenced how and why technology was and is developed. SST was a departure from earlier research traditions which viewed social adjustments solely as a reaction to technological change, according to Williams and Edge (1996). While SST was a broad category of individual theories designed to understand how society shaped and interacted

with technology, the emergent leader of these theories was the social construction of technology theory.

Social Construction of Technology

In their theory of social construction of technology (SCOT), Pinch and Bijker (1984) asserted that technology was open to sociological analysis, both of its use and its design. According to Pinch and Bijker, technology was successful because it was accepted by a social group, a concept they referred to as interpretive flexibility. Bijker (1997) later went on to add the idea of technological frames as a means to contextualize how members of these social groups interacted with each other and what elements influence those interactions. Examples of these elements include goals, organizational constraints, or tacit knowledge.

The concept of social acceptance is essential to the idea of women in computing (computing being a technological derivative) because, according to Wajcman (2000), women were mostly absent from the groups responsible for designing technology and therefore not much thought was given to what elements of technology women might find necessary or even attractive. Wajcman also pointed out that when the scope of observation of the technology development process was widened to all related processes, such as marketing, sales, consumption and end-use, the relevancy of female input to the process increases exponentially. The lack of input by or consideration of women in the technology design process contributed to a lack of social identification of women seeking technology-related jobs.

Social Identity Theory

Tajfel and Turner (1979) presented the social identity theory (SIT) as an alternative to the explanation of intergroup behavior using functional relations between social groups by Sherif (1966). The core of SIT, according to Tajfel and Turner, was that people stereotyped due to a natural cognitive desire to group things. SIT presented the dichotomy of in-groups (of which the

individual was a member) and out-groups (Of which the individual was not a member) and stressed the differences between the out-group while highlighting the similarities of members of the in-group. According to SIT, three factors contribute to the in-group/out-group mindset: social categorization, social identification, and social comparison.

Social categorization. According to Tajfel and Turner (1979), social categorization is a means by which people understand and identify objects. Social categorization works by placing a social lens on the situation, allowing for people to be categorized within a social environment, for example, by race, gender, political affiliation, or occupation. Tajfel and Turner also asserted that individuals could learn about themselves and define appropriate behavior based on the social categories to which they felt they belonged.

Social identification. After achieving social categorization, social identification is the process of assuming the identity of the group or groups of which we classify ourselves, according to Tajfel and Turner (1979). Establishing an identity with a group leads to the individual modeling behaviors based on their perception of how a member of the group or groups to which they belong should act. Tajfel and Turner also linked self-esteem to group membership.

Social comparison. The last level of the SIT is that of social comparison. With self-esteem linked to a social group, Tajfel and Turner (1979) argue that the groups that we are a member of must favorably compare with other groups. Failure to do so creates a rivalry, which Tajfel and Turner assert is a cause of competition resulting from the need to maintain self-esteem. According to Tajfel and Turner, understanding the importance of social comparison is critical to understanding prejudice.

Social Mobility and Social Change

Tajfel and Turner (1979) discussed two more elements of SIT, in regards to group members that are especially relevant to the topic of gender diversity in computing. The first element is that of social change. According to Tajfel and Turner, the concept of social change is related to an individual's inability to ascribe to or place themselves as a member of a group that they deemed to be unsatisfactory, underprivileged or socially stigmatized. An example of this would be a woman not seeking education or a career in computing based on her perceived misalignment with the members of that profession. Likewise, Tajfel and Turner present the concept of social mobility, which argues that individuals can move between social groups as they begin to feel less aligned with the ideals or tenants of said group. An example of this would be a woman leaving a job in computing based on her perceived misalignment with the members of that profession.

Gender Gap and Diversity

From a purely descriptive statistical viewpoint, it is possible to infer the effect of the social identification element of SIT, or the lack thereof, for women in the computing industry. According to the US Department of Labor Bureau of Labor Statistics (2017), in 2017 the number total number of working Americans was 157 million. Women represented approximately 47% of the workforce. Even though women comprised almost half of the overall workforce, a 2013 study showed that only 26% of computing industry workers were women (National Center for Women in Technology, 2014). In 2017, women comprised only 25.5 % of the 4.8 million computing professionals (Bureau of Labor Statistics, 2017), which indicates that the number had continued its decline. The disparity in gender was not only alarming, but it also begged the question of what factors contribute to the skewed numbers and whether or not those factors could equally affect other industries.

The period between 1990 and 2013 showed the most significant decline of women in computing, with one of the most prominent factors being the release of the personal computer and the beginning of the idea that computers were for boys, according to Henn (2014). Of interest was that during the same period, from 1990 onward, the number of total available jobs in the computing industry had grown 338 % according to Funk and Parker (2018a). Not only was the number of women seeking jobs shrinking, but the total available number of jobs was increasing, further widening the gender gap in the industry.

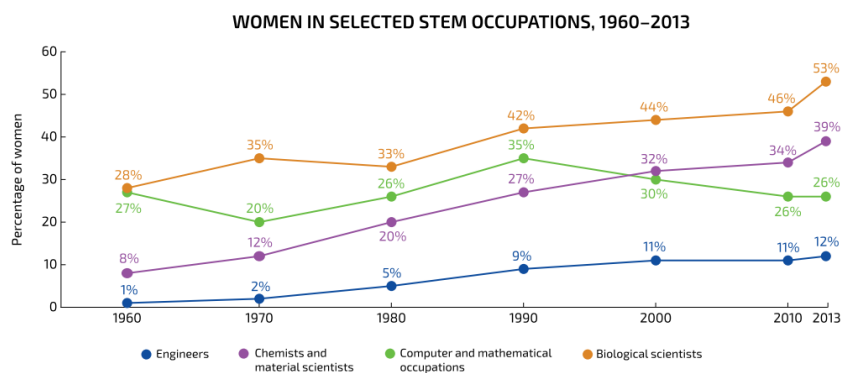
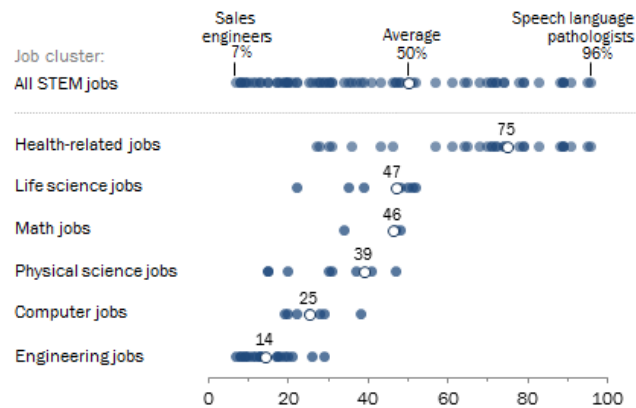


Figure 1. Women in Select STEM Occupations, 1960 – 2013. Reprinted from *Solving the Equation* (p. 9) by C. Corbett and C. Hill, 2015, Washington, DC: AAUW. Copyright 2015 by the American Association of University Women. Reprinted with Permission.

Further research conducted by the Pew Research Center showed that the trend continued between 2014 and 2016, with women comprising only 25% of the computing workforce. Figure 1 illustrates the juxtaposition between the decline of women in computing disciplines, compared to women in other STEM disciplines. Figure 2 shows the number of women in STEM disciplines from 2014 to 2016.

Representation of women in STEM jobs varies widely

Share of employed adults in each occupational group who are women (%)



Note: Based on employed adults ages 25 and older. Each circle represents a single occupation (e.g., mechanical engineer, registered nurse). Engineering includes architects. STEM stands for science, technology, engineering and math.
Source: Pew Research Center analysis of 2014-2016 American Community Survey (IPUMS).
"Women and Men in STEM Often at Odds Over Workplace Equity"

Figure 2. Representation of women in STEM jobs from 2014 to 2016. Reprinted from Pew Research Center, Retrieved August 8, 2018, from http://www.pewresearch.org/fact-tank/2018/01/09/7-facts-about-the-stem-workforce/ft_18-01-08_stemworkers_5/. Copyright 2018 by Pew Research Center. Used with permission.

Of the numerous contributing factors, gender inequality is that of fair compensation. According to the American Association of University Women (AAUW) (2014), in 2012 women earned 25% less pay than their male counterparts performing the same work. The AAUW research indicated that job choice played a factor in pay equality, as traditionally male-associated professions, such as computing, have higher pay rates. According to Funk and Parker (2018a), the average woman working in a STEM occupation made two-thirds more than one working in a non-STEM occupation. Further, Funk and Parker found that women who majored in a STEM

occupation in college out-earned women who did not. The opportunity for a high paying job poses the interesting question of why fewer women pursued careers in computing if the pay was attractive.

Benefits of Diversity

Numerous studies discovered the benefits of having a diverse workforce. Nelson (2014) argued that gender, social, and cultural diversity in teams led to more innovation and improved financial results over homogenous groups. Additionally, Funk and Parker (2018b) pointed out that in a 2017 survey 78% of adults in the U.S. felt that gender diversity was important in the workplace. Of the respondents who identified gender diversity as necessary, 52% recognized that it was essential and 26% determined that it was somewhat important.

Benefits of a diverse executive board. According to Grey and Cox (2016), women serving on executive boards led to increased organizational performance. The researchers also pointed out that these companies reported higher profits because having a gender homogenous executive team increased cognitive bias and led to the improper allocation of resources. Hersh (2016) argued that gender-diverse executive boards could be critical to an organization's success because of their unique experiences and perspectives. Hersh pointed out that failing to include women on executive boards limited those experiences and perspectives. She also echoed the opinions of Nelson (2014) in that executive boards that contain women lead to increased financial success.

Reinforcing the claims made by Nelson and Hersh, Sonnabend et al. (2017) reported that in 2017, women held more than 20% of the executive board seats of *Fortune 1000* companies. Sonnabend et al. (2017) also found that more successful companies had a higher percentage of female executive board members, with *Fortune 100* companies averaging over 24% of their members being women.

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According to Collamer (2017), being selected to be a member of an executive board was extremely difficult, especially in the case of for-profit organizations. Being a competitive candidate required a significant amount of knowledge and experience in the industry, which presented the issue of a smaller pool of female candidates with the extensive computing experience needed.

Impact on engagement and retention. Along with profitability, a diverse gender workforce had a positive effect on employee engagement and retention, according to Derven (2014). Derven found this to be the case for those organizations that offered employee resource groups (ERGs). ERGs are affinity groups that help to facilitate a more welcoming environment, especially amongst newer employees. Hui and Farnham (2016) pointed out that while developing an organizational culture that includes diversity, groups such as ERGs are critical to provided minority groups a sense of professional and emotional support. Additionally, Jansen, Otten, and Zee (2017) warned against the effect of a perceived negative gender diversity climate on absenteeism. According to their study, women who felt that their organization was not gender-inclusive were prone to higher rates of absenteeism.

Implicit Bias

The underlying theme of implicit bias in computing gender diversity was one that warranted much more research, according to Buse, Hill, and Benson (2017). Women faced unconscious biases about technical proficiency and capabilities that were not experienced by their male counterparts. To help combat these biases, leaders should encourage activities that highlight and increase these proficiencies and skills. Additionally support for women to attend training, conferences, and other events will not only help improve their technological skillset but also instill confidence in their abilities – making women feel more comfortable in demonstrating

their expertise. It should also serve to lend a level of credibility amongst peers that can combat any biases, unconscious or otherwise (Anita Borg Institute, 2013).

Barriers to Entry for Women Seeking Computer Science Education

Researchers have argued that the lack of women pursuing STEM degrees is increasing. According to a National Science Foundation (2015) study, for example, women earned 57% of all bachelor's degrees and nearly half of all science and engineering degrees since the late 1990s. These results were consistent with the rise in STEM occupations, except computing, as seen in Figures 1 and 2. While the number of women obtaining degrees increased over time, the percentage of the population of female recipients decreased most significantly for undergraduate degrees between 2002 and 2012.

The concept of self-alignment with a perceived stereotype was a recurring theme when discussing gender diversity in computing. It was not surprising to see the perceived stereotype issue present when talking about deterrents to women seeking Computer Science (CS) degrees. The deterrence may have occurred long before women chose to pursue secondary education.

Schimpf, Andronicus, and Main (2015) argued that there were impediments that prevented a balanced gender use of computers in the early educational system. These impediments changed as the student transitioned from primary school to the collegiate level, but at each phase, they indicate a deterrent towards computer use in women. According to Schimpf et al.,- (2017) at younger ages, research showed that boys tended to monopolize computer use, while at higher levels, girls were marginalized and struggled to identify with the computer science student stereotype.

Perceived Stereotype and Alignment

Govender and Khumalo (2014) observed that one of the greatest influences on a woman's decision to pursue a degree in Information Systems (IS) was the level of difficulty women

expected from the course of study and confidence in their ability to complete the required coursework. Misconceptions were especially detrimental to those minority students (race and gender) that were further away from the perceived CS stereotype of a singularly focused, competitive, asocial, male (Lewis, Anderson, & Yasuhara, 2016).

Research into the relationship between gender stereotypes and interest in science as a career has shown that there was a direct relationship between self-identification with stereotype and career aspiration, which clearly illustrated the SIT concepts of in-groups and out-groups. More specifically, the inability to self-identify with the stereotype of a computer science student prevented this group from becoming an in-group, which caused computer science students to become an out-group for many women. Cundiff, Vescio, Loken, and Lo (2013) found that the stereotype of men perceived to be more science-oriented led to lower levels of female interest in science. Cundiff et al. based this finding on a study of roughly 1800 STEM students. While this sample size was small, it was representative of the previous conclusions by Nosek and Smyth (2011) and even the early theories of Lewin (1951). Additionally, lack of identity alignment reduced female interest in pursuing a career in science. The same stereotype reinforced male interest in science and interest in career aspirations in the science industry (Cundiff et al., 2013).

Bock, Taylor, Phillips, and Sun (2013) conducted a follow-up study to augment existing studies on why individuals decided against pursuing a degree in CS. Their findings reinforced initial results that indicated that incoming first-year students were not interested in CS degrees. Specifically, women reported that they felt that CS degrees were designed to support males better. Other minorities said their lack of exposure to CS was an inhibitor. Minorities also stated that they thought that CS degrees were not a fit due to their perception of the CS student stereotype.

The real danger in stereotypes was that they were still influential, despite being found untrue. According to Hill, Corbett, and St Rose (2010), negative stereotypes regarding women's and girl's abilities in math and science persisted, even though that gap had closed. Hill et al. (2010) found that this phenomenon, referred to as stereotype threat, actually caused women to perform at lower levels than their male counterparts, when present. The findings by Hill et al. (2010) corroborated a 1999 study on the influence of stereotype threat on women's math performance, conducted by Spencer, Steele, and Quinn, (1999) which showed that math performance was roughly equal when stereotype threat was not present.

Ironically, in a study conducted by Cheryan, Plaut, Handron, and Hudson (2013), which tested the influence of participant perception of CS stereotypes on their willingness to choose CS as a major, participants viewed false media reports that contradicted accepted stereotypes. Specifically, these reports presented the idea that the levels of male and female CS students were closer to being equal and that more women were pursuing CS degrees. Results indicated that while contradictory media did not influence the choice of male participants, female participants reported that they would be more willing to pursue a CS degree, reinforcing the importance of identity alignment with perceived stereotypes for women considering CS degrees, and illustrating the effect of in-group alignment.

Self-efficacy. Closely linked to the topic of perceived alignment with a stereotype is the concept of self-efficacy, or the belief in one's own ability to succeed in a specific situation (Bandura, 1997). There was a significant amount of literature that discussed how self-efficacy, or the lack thereof, was a tremendous deterrent for women that may have had an interest in pursuing a career in CS. Heilbronner (2013) suggested that family, teachers, and other mentors play a critical role in helping young women to develop a strong sense of self-efficacy. In a study

conducted by Beyer (2014), she found that women with low self-efficacy in respects to their skill with computers were less likely even to take a CS course, let alone choose to pursue a degree.

The danger in self-efficacy lies in its ability to act as a deterrent from taking CS courses. Lin (2016) found that male participants in her study self-reported higher levels of self-efficacy in learning, computer, and programming categories. The results of the study, however, did not significantly determine that their ability to perform in those categories was any higher than their female counterparts.

Exposure to fundamental concepts. Several studies have pointed out that varying levels of exposure to fundamental computing concepts can have a significant impact on the desire to pursue and obtain a degree in computing. Berg, Apel, and Leicht-Scholten (2014) found that there were substantial gaps in female student exposure to key concepts that created educational obstacles. These gaps have the potential to become a significant deterrent when choosing a degree program. Similarly, Cohoon, Cohoon, and Soffa (2013) found that women and other underrepresented minorities in introductory CS courses were likely to have a more varied level of exposure to computing concepts than their peers—ranging from very little exposure to years of instruction. They also attributed the gender differences to negative cultural stereotypes about gender proficiency in computing, which can lead to lack of confidence and a reduced sense of belonging.

However, Hill, Corbett, and St Rose (2010) argued that the belief that men and women have differing aptitudes for mathematics and language skills based on the biological differences between genders is untrue. Using this argument as an explanation for why there are higher numbers of men STEM has fallen out of favor. Recent improvements in math skills for girls contradicts this postulation, according to Hill et al. (2010). This improvement also indicates that

positive encouragement for girls to improve STEM core competencies may lead to higher numbers of women interested in pursuing an education, and ultimately, a career in STEM fields (Hill, Corbett, & St Rose, 2010).

In a Stanford University study, Redmond, Evans, and Sahami (2013) found that starting CS classes earlier in the undergraduate period of their education was a critical element in generating a higher number of female CS majors. Additionally, the study reinforced findings from similar studies that female students may have a different history of exposure to the CS field than their male peers. Redmond et al. (2013) recommended that each student's exposure level be determined, and actions are taken to proactively engage students that may not have had support in past CS classes.

Early Exposure

Even before the collegiate level, higher levels of exposure to computing concepts are critical to fostering and continuing women's interest in computing, according to Brady et al. (2017). Although their study included a minimal sample size of just thirteen girls, Brady et al. (2017) found that targeted workshops for women help to align current interests with computing concepts. Coincidentally, in a study designed to identify characteristics of women succeeding in a CS course of study at both public and private technology institutions, Brandt (2014) found that most of these women had committed to a STEM career by the tenth grade, which further emphasizes the importance of early exposure. Brandt's research reinforced similar findings by Marcu et al. (2010) in their paper that discussed the design and evaluation of a CS and Engineering course for middle school girls, in that both situations served to introduce or reinforce interest in further CS courses by participants. Engagement of women before choosing a collegiate major is critical because McGill, Decker, and Settle (2016) found that women who positively viewed STEM outreach programs were more likely to pursue a STEM major. Proof of

this success is observable in the increased participation by female students in Advanced Placement (AP) CS exams, according to Partovi (2017), who reported increased participation of 135% by female students between 2016 and 2017.

Creating engaging learning environments for girls seeking to learn more about computing is essential to the success of those initiatives. Medel and Pournaghshband (2017) found that many instructional CS materials projected unfair gender biases. They advised that these materials be adjusted to suit the needs of all students, not just one specific gender. Frieze and Quesenberry (2013) warned against altering the environment to meet gender-specific likes and dislikes of one gender. By adopting this Essentialist point of view, one in which men and women view things differently based on their gender, we are reinforcing gender stereotypes. Rather than creating a common environment where both men and women are comfortable and can operate successfully on an equal level, this approach creates a more significant gap. This approach also risks alienating men because the environment no longer meets their gender-specific needs – effectively shifting the gender imbalance in the opposite direction.

Lack of role models. Exacerbating the issue of identity alignment is the perceived lack of female role models and mentors. Schein (1975) was perhaps one of the first to call attention to the fact that gender stereotyping was a contributing factor to the barrier that women face when seeking to attain leadership positions. Vardi (2015) argued that it was imperative that faculty members of both genders commit to a mentoring program that solidified relationships with students and fostered an interest in computing activities outside the classroom, such as research. Berg et al. (2014) reported that the existence and availability of positive female role models were likely to increase the interest of women in CS degree programs. They proposed that increasing the exposure of female CS candidates to successful women in the career field, through media

exposure or school visits, would be critical in both increasing interest and breaking down the perceived stereotype.

In a similar study, Denner, Werner, O'Connor, and Glassman (2014) found that women reported less support and encouragement from classmates to pursue CS studies, which presented a danger because this encouragement represented one of the primary predictors of intent to major in CS. Denner et al. (2014) found that the importance of the effect of peer support diminished over time, further highlighting the importance of positive encouragement early on in the pursuit of a CS degree.

Alvarado and Judson (2014) analyzed the survey responses from attendees of an annual conference targeting women for recruitment into CS academic programs and found indications that conference attendance may help to overcome common roadblocks facing women considering CS as a major. Alvarado and Judson (2014) argued that being exposed to real-world CS demonstrations, discussions, and women who have successful industry careers goes a long way towards overcoming preconceived notions of the stereotypical asocial, competitive, white male CS practitioner. It also served to reinvigorate the ambitions of current CS majors that were wavering in their commitment to the program as well as undecided or previously uninterested students.

Stout et al. (2016) conducted a study on the effects of membership in a grad cohort for Ph.D. students on retention. Stout et al. found that in this context, peers acted in a similar role as professional and academic role models, increasing retention rates. Additionally, these students had a higher desire to become well-known in their career and give back to the community.

Lastly, Vitores and Gil-Juárez (2016) argued that too much of the literature surrounding gender diversity in computing focuses on what is referred to as the “leaky pipeline” (p. 5) model

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in which loses potential candidates for CS degrees and computing careers along the way. They presented the alternative of highlighting the achievements and contributions that women have made to the industry. By doing this, Vitores and Gil-Juárez (2016) argue that positive role-models will be more evident to women seeking examples of success.

Barriers to Entry for Women Entering the Computing Industry

The gender gap not only reflects itself in computer science degree programs but also in the industry. According to Graf et al. (2018) women comprised roughly 25% of American computing industry workers, as seen in Figure 2. That percentage is not surprising when considering that the rate of women awarded computer science degrees in 2012 was less than 20%, according to the National Science Foundation (2015).

Researchers studying the gender gap have developed various theories as to why there was a shortage of women entering computing professions. Klein and Díaz-Hernández (2014) postulated that a hidden bias on the part of hiring managers might have had some impact on the number of women entering the workforce. According to their theory, due to perceived stereotypes regarding what constitutes a computing professional (in general, a Caucasian or Asian male). This bias was consistent with the same gender-biased issues that occurred in the Academic world. Klein and Díaz-Hernández (2014) argue that the human brain tends to recognize patterns and subconsciously rejects those items that do not fit that pattern. Hence an implicit bias towards a male candidate in most cases. Gamm (2014) presented the importance of being introduced to and inspired to pursue STEM careers by influential people in their lives, such as parents, teachers, and mentors.

Regardless of the plausibility of that theory, as many as 45% of women entering the computing profession leave in the first five years. Compared to a 17% attrition rate amongst men

this was indicative of a problem (Barr, 2017). Barr argues that merely attracting more women to the profession has done very little to change the climate to one that is more gender accepting.

According to Shields (2015), the issues facing the women in the computing industry did not just lie within the educational inroads that precede this cycle. There was also present a disenfranchisement of women within the industrial cycle causing them to leave the system entirely. The departure of women further diminishes the already lopsided gender balance; therefore it is imperative that the industry does its best to retain female talent to ensure gender diversity.

One common theme regarding retention revolves around the concept of gender-based roles regarding familial responsibility. Beasley, Lomo-David, and Seubert (2001) asserted that it is imperative that IT leaders have an understanding of the difference between the gender-based perception of professional and domestic roles. Beasley et al. (2001) contended that IT leaders must learn how to leverage technology to support women who still want to have a traditional domestic role, yet do not want to relinquish their professional responsibilities. The importance of supporting the needs of the entire workforce is even more critical in the US now, with a shortage of skilled computing professionals.

Personality Traits and Advancement

As far back as 1979, Brenner and Grenhaus found that attributes considered critical to managerial success, such as aggressiveness and dominance, were deemed to be male-oriented attributes. Additionally, Hua (2010) suggested that the presence of strong self-confidence and ambition were critical to a long-term career for women in the computing industry. While those qualities may be necessary for men in the industry as well, they were even more critical for women who faced increased barriers to entry and promotion through the ranks. Williams (2014) pointed out the irony that when women displayed behaviors that were considered vital elements

of what make men successful – assertiveness, competitiveness, aggressiveness – they were criticized, ostracized, even threatened in today’s computing industry. Behavior such as this contributed to an unwelcoming climate that has seen an exodus of women leaving the industry, with 41% of women leaving jobs in computing after ten years. Since 1985, there was a 10% decline in the number of women holding jobs. The decline reflected the findings of Hackett and Betz (1981), which were that both internal (self) and external (management) expectations of low performance were detrimental to career progression. Kossek, Su, and Wu (2017) argued that traditionally initiatives to stem the loss of women focused on changing attributes of women, rather than changing the context that caused disadvantage towards women. That approach addressed a symptom, rather than a problem.

Lack of Women in Leadership Roles

Fine, Wendt and Carnes (2014) presented additional factors that contributed to the loss of females in the industry. Critical among these factors was the underrepresentation of women in senior leadership positions, which added to a tiny pool of role-models and mentors. Claggett (2016) also argued that gender barriers contributed to the low number of women that attain senior leadership positions in the computing industry, which are critical to correcting the gender imbalance. Additionally, differing gendered expectations of women and lack of career development and work-life balance support caused women to seek careers in industries that were more supportive in these areas (Fine et al., 2014).

Perceived Stereotype of STEM Workers

There is much literature that discusses the impact of perceived stereotype and self-perception and alignment with those stereotypes among women considering the computing profession. While this phenomenon may have had a more substantial effect several decades ago, some more recent studies showed that those women who did choose to pursue a computing

profession largely ignored those stereotypes, according to Faulkner and Lie (2007). However, they did acknowledge that campaigns designed to promote women in computing did have a positive effect on changing the perceived image of women in the computing industry.

Stereotype threat. DuBow (2013) warned of the dangers of stereotype threat. This phenomenon tended to occur in male-dominated classrooms or offices and was a self-inflicted occurrence. According to DuBow (2013), when an active stereotype that women do not perform as well in technology was present, it caused women to perform worse due to unconscious bias. In a related study, Holleran, Whitehead, Schmader, and Mehl (2011) found that mixed-sex conversation led female coworkers to increased disengagement from their work. Conversely, same-sex conversation amongst male coworkers led to increased engagement and collaboration.

In a recent study on how gender stereotypes influence the people's justification of gender inequalities in the workplace, Cundiff and Vescio (2016) identified a direct link between reinforced stereotypes and gender disparity justification. Their research found that individuals that prescribed to stereotypical gender roles were more likely to discount gender discrimination as a factor in workplace disparities. They argued that failure to acknowledge gender-based discrimination recognition legitimized the inequality. Additionally, Karpowitz, Mendelberg, and Shaker (2012) explained that according to their "gender role" hypothesis (p. 534) women have lowered status and are less participatory when they represent the minority population of a group. The greater management community should be aware of this phenomenon because according to Taylor (1914) of equal importance to employer success, is employee success.

Women in Computing Outside the United States

The preponderance of literature on the topic of women in computing, as reflected so far, has been from studies of women in the US. It would be remiss to not at least touch on some of the literature that focused on the phenomenon of gender imbalance in the computing industry in

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other countries. The following research shows that the phenomenon was not specific to the American culture. However, some countries and cultures did not display the same stereotypes regarding women in computing.

Cultural Differences

There is some literature that discusses the cultural impact of women seeking education and careers in the computing industry. Schinzel (2002) pointed out the issues present in developed countries like the U.S. and Great Britain, such as the declining number of women participating in CS courses, did not occur in semi-developed countries like India, South America, and Africa. The phenomenon indicates that there may be factors unique to more developed countries that create barriers to entry for women.

Germany. The difference between developed and semi-developed cultures was especially visible, according to Schinzel (2002), when viewing the country of Germany. Schinzel asserts that before the fall of the Berlin Wall, less developed East Germany had an equal distribution of gender participation while more developed West Germany had much lower levels of female participation. After the fall of the wall, the former East German countries saw a decline in gender diversity, settling at comparable levels to that of former West Germany.

Ireland. In a study conducted in Ireland Mackeogh & Preston (2004), focused on the governmental category of late adopters, in which women were a subset. Their study presented how an environment that supported the needs of both male and female computing professionals helped to meet the needs of a growing tech economy. Interestingly, Mackeogh & Preston (2004) found that there was considerable interest in computing in lower socioeconomic classes, especially with women, when the gender inclusive programs were made available.

India. In an Indian study, Gupta (2014), named patriarchal motives and a positive role of women in computing professions as motivators for young women to seek education and

employment in computing. As with other countries, Gupta (2014) found that economic and socio-cultural issues played a role in influencing women's desire to pursue education and careers in computing. Gupta (2014) also indicated that the strong familial focus of the Indian culture and Patriarchal motivation were great motivators for women to pursue an education in computing fields.

Malaysia. According to Lageson (2008), Malaysia provided a compelling case for gender diversity in computing. According to a 2005 study conducted by Lageson, women comprised about half of the CS degree recipients in Malaysia. Lageson credited this to a difference in the way that women perceived computing jobs, which is supported by a Government sponsored program to increase information and communication technologies capabilities. Additionally, Lageson pointed out that the expanded role of women in religious and political offices may have supported the mindset that traditional gender stereotypes in all areas were no longer relevant.

Mexico. De Lara and Liu (2016) argued that in Mexico non-wealthy women have very little access to CS education and employment opportunities. Additionally, De Lara and Liu (2016) pointed out that Mexican culture is very male-dominated and that the association of computing with masculine attributes creates another barrier to entry for women. However, recent tech growth in the Cali-Baja Mega-Region, which is a bi-national initiative between the US, primarily San Diego, and the Mexican Tijuana/Baja Peninsula business to become a technology epicenter that rivals Silicon Valley (Cali-Baja, n.d.). A once tourism-based economy had become a burgeoning hub for cheap, yet skilled computing resources, with talented workers that cut their teeth in this new environment before moving on to more traditional technology organizations like Google.

Identifying the Literature Gap

While reviewing the literature surrounding the concept of gender diversity in computing, it was quickly evident that the preponderance of research used the quantitative research method. Of the 83 artifacts reviewed, over half (46) used a quantitative research method. The next largest category was aggregates (21), or artifacts that reviewed multiple other studies, with at least half of those being quantitative studies. Four studies used a mixed method approach, all focusing on women seeking computer science education. Of the 13 qualitative studies, four were from countries other than the US, two were not exploratory, five did not focus specifically on women in computing, and one looked only at women seeking community college education in computing.

Of the 13 reviewed qualitative studies, the closest artifacts to this dissertation were both very similar (Walters, 2015; Claggett, 2016), both of which focused on gender-based barriers that women faced when seeking leadership positions. The slight nuance being that Claggett (2016) discussed barriers at a general leadership level and Walters concentrated on STEM leadership. What remains is a gap in the literature that solicits direct input from women in the computing industry as to how their experience has allowed them to overcome or mitigate the identified barriers to entry, as well as let them remain in the industry successfully.

Literature Gap

The gap in the literature that the proposed research seeks to address is the lack of direct input from women that have successfully obtain education and employment in computing as to how they successfully overcame or mitigated the identified barriers to entry, as well as allowed them to remain in the industry successfully. The gap is relevant because it looks at a different segment from the numerous studies that have focused on getting more women into computing education (Alvarado & Judson, 2014; Beyer, 2014; Cohoon, Cohoon, & Soffa, 2014; Medel &

Pournaghshband, 2017). Additionally, it addresses Vitores and Gil-Juarez' (2016) criticism of the current literature's pessimistic view of the "Leaky Pipeline" (p. 1) metaphor, which is used to describe the barriers to entry approach favored by most researchers.

To be fair, those same barriers are presented in this literature review, but being a review, it only speaks to the nature of the literature. Presenting those barriers provides the opportunity for the proposed research to focus on the positive experiences of women that attracted them to the computing industry and allowed them to not only be successful in the industry but also to persist, despite the fact the identified barriers to entry are still present.

Conceptual Framework

There has been much research already conducted the topic of gender diversity in STEM disciplines (Achiam & Holmegaard, 2016; Barr, 2017; Buse et al., 2017; Corbett & Hill, 2015; DuBow, Quinn, Townsend, Robinson, & Barr, 2016). The main area of study has been on why more women have not been seeking CS degrees. (Blakley, 2016; Cohoon et al., 2013; Denner et al., 2014; DuBow et al., 2016; Falkner, Szabo, Michell, Szorenyi, & Thyer, 2015). Equally studied were ways to retain female talent once in the industry (Corbett & Hill, 2015; Holtzblatt, Balakrishnan, Effner, Rhodes, & Tuan, 2016; Lewis et al., 2016; Walker, 2015; G. Williams, 2014).

The four-layer conceptual framework presented in Figure 3 shows barriers that prevent women interested in computing from attaining a career in computing and is based upon the Expanded Social Cognitive Career Theory (SCCT) Model for Attrition/Persistence (Singh & Fouad, 2011). SCCT, according to Singh and Fouad, can be used to predict how relationships between self-efficacy, expectations, and goals and uses that relationship can affect career development. The expanded SCCT model presented by Lent et al. (2003) introduces the

elements of barriers and supports, which directly links barriers to career choice through self-efficacy.

The topmost layer in the conceptual framework represents women who are interested in computing as a profession. While there have been no studies that identify the total number of women who are seeking jobs in the computing industry, women make up almost half of the total workforce in the United States (English, Hartmann, & Hayes, 2010).

The second layer represents the educational barriers that women face. At this layer, self-identity alignment and competition for access to resources represent the primary barriers (Alvarado & Judson, 2014; Beyer, 2014; Bock, Taylor, Phillips, & Sun, 2013; Cundiff & Vescio, 2016; Lewis, Anderson, & Yasuhara, 2016). Gamm (2014) presents the importance of being introduced to and inspired to pursue STEM careers by influential people in their lives, such as parents, teachers, and mentors.

The next layer represents workplace barriers. Those women who do obtain the required educational background for a computing career still face significant obstacles. At this layer self-identity, alignment issues persist and accompany unconscious bias by male counterparts, as well as a gender-based double standard. Klein and Díaz-Hernández (2014) postulated that a hidden bias on the part of hiring managers might have some impact on the number of women entering the workforce. According to their theory, due to perceived stereotypes regarding what constitutes a computing professional (in general, a Caucasian or Asian male). Klein and Díaz-Hernández argued that the human brain tends to recognize patterns and subconsciously rejects those items that do not fit that pattern. Hence an implicit bias towards a male candidate in most cases.

The final layer represents the smaller group of women who have successfully achieved a career in computing. Based on the 2013 US census data, this represents only 26% of the

computing jobs in the US – down from 34% in 1990 (Corbett & Hill, 2015). Retention is still a significant issue once women successfully obtain jobs in the computing industry. Barr (2017) reported that as many as 45% of women entering the computing profession leave in the first five years. Compared to a 17% attrition rate amongst men this is indicative of a problem. Barr (2017) argued that merely attracting more women to the profession has done very little to change the climate to one that is more gender accepting.

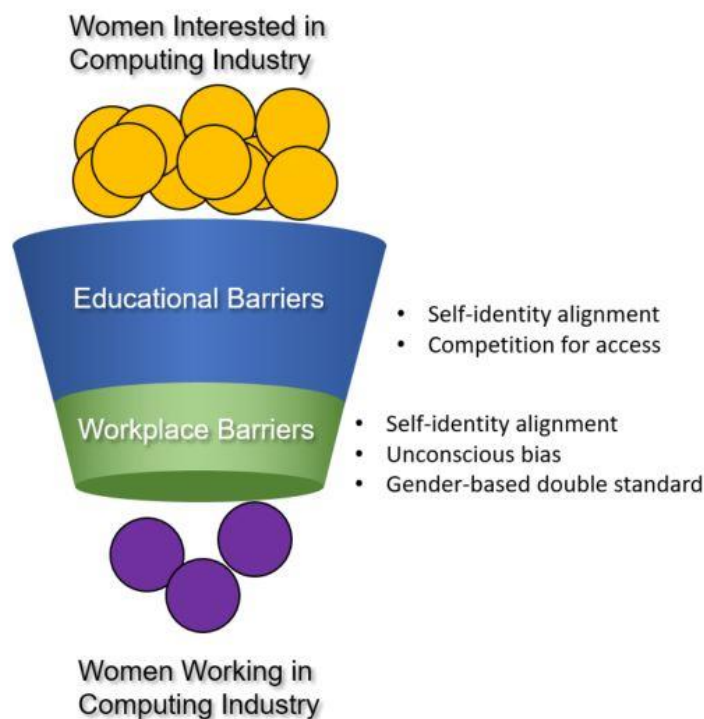


Figure 3. Barriers to Entry for Women Seeking Computing Jobs

Summary of Literature Review

The literature regarding gender diversity in computing is quite extensive. Despite concerted efforts in understanding why this phenomenon has occurred, there continues to be a decline in the number of women in the computing industry (NCWIT, 2014). There seem to be two major drop-off points for women seeking a computing career, the first being when pursuing a degree in computer science, the second being after graduation while attempting to gain entry to and remain within the industry.

Despite the two very distinct drop-off points, there exist several common factors that act as barriers to entry. The first factor being that of perceived stereotype and identity alignment. Research supported the phenomenon that women who fail to see themselves as a stereotypical computer science student or computing professional either chose not to pursue those goals or failed to achieve those goals (Cundiff & Vescio, 2016; Govender & Khumalo, 2014). The second common factor acting as a barrier to entry was the lack of female role models, with Schein (1975) being one of the first advocates for the need to increase female role models in the industry over forty years ago. Lastly, and perhaps the most disturbing of the common factors acting as a deterrent is that of implicit bias or the subconscious belief that gender influences the ability to be successful (Buse et al., 2017). Buse et al. (2017) argued that much research is still needed to understand the entirety of the impact that implicit bias plays in preventing women from pursuing education and careers in the computing industry. Based on the literature reviewed and the gap identified, a qualitative exploratory approach will be the research method used for this dissertation. The purpose of that research is the identification of positive factors experienced by women in the computing industry that have helped to overcome, or at least mitigate, these common barriers. The next chapter discusses the conduct of the study, to include the population selection and sampling procedure, as well as how the data will be analyzed.

CHAPTER THREE

The problem addressed in the proposed study was that strategies technology managers need to retain women in the computing industry have not been identified (Beyer, 2014). The purpose of the qualitative exploratory study was to explore the strategies technology managers need to retain women in the computing industry. This chapter discusses the method of research used to conduct the proposed study. Discussion on the data collection process provides details about the target population and sample, as well as how that data was analyzed. Lastly, we will examine information about human subject protection and other ethical considerations.

Research Tradition

A research *approach* is a path that a researcher takes to get from assumptions regarding a specific topic, through data collection, and culminates with the analysis and interpretation of the collected data according to Creswell (2014). Additionally, Creswell pointed out that a research *design* is the specific method of inquiry in which a researcher plans use when gathering data. Furthermore, he asserts that a research *methodology* is a manner in which a researcher goes about collecting information about a specific topic. Additionally, Kallet (2004) asserted that a methodology is used in research to apply the appropriate techniques required to collect, process, and analyze data in an effort to allow a reader of the study to ascertain its reliability and validity.

The qualitative methodology was used for the study because, according to Creswell (2014), it seeks to understand phenomena and to explore and understand the factors around that phenomena. The quantitative methodology was not used for the study because existing quantitative research has already reviewed the phenomena, such as the study conducted by Corbett and Hill (2013), which used US Census data to provide a quantitative representation of the low number of women in computing. The mixed methods methodology was not feasible for the study due to time limitations.

A research design is organized by population and sample, sampling procedure, instrumentation, validity, reliability, data collection, data analysis, and ethical considerations. Research, at its core, depends upon several ontological (regarding the nature of reality) and epistemological (regarding the nature of knowledge) assumptions, according to Dayton (2011). Dayton (2011) argued that people assess reality in two ways. Reality can be assessed objectively, separate from an observer, which allows for an equally objective quantification and measurement. Alternatively, we view reality as the specific interpretation of individual participants. The latter form of assessment allows for a deeper understanding of events, which is especially relevant to the proposed research topic.

Goertz and Mahoney (2012) pointed out that the means by which quantitative and qualitative scholars view concepts and measurement differ vastly and tie tightly to ontology and epistemology, with qualitative researchers using semantic processes to identify attributes that provide meaning to concepts, while quantitative researchers construct concepts by factors that (potentially) cause or are caused by a concept. From an epistemological viewpoint, qualitative researchers feel most confident about their research when dealing with extreme values, while quantitative researchers have high confidence in middle values, according to Goertz and Mahoney (2012).

Based on the selection of the qualitative methodology, an exploratory design approach was used because, as Mansourian (2008) argued, the benefits of using exploratory research when dealing with ambiguity in a subject or when there are uncertain elements within the research topic. Mansourian (2008) relates exploratory research to Glaser and Strauss' (1967) process of theory emergence that arose from the grounded theory research methodology. Based on the current literature surrounding the topic of women in computing, paired with the continued

decline of women pursuing careers in the industry, it is easy to make the argument that there is definite uncertainty, as well as potential ambiguity relating to the topic. While there is no shortage of theories or hypotheses as to why women are deterred from seeking education and employment in the computing industry, the continued decline represented in the data suggests that a means to correct this decline has not been found or implemented by current technology managers (Corbett & Hill, 2015).

Three other design options were considered: ethnography, grounded theory, and phenomenology. The ethnographical design was not feasible for the study because of the time and physical commitment required by the researcher regarding embedding within the population. The grounded theory design was not used for the research because the study seeks only to identify relevant conditions or phenomenon. Phenomenology would be a good option, as it focuses on the lived experiences of the participants who have experienced a shared event and used interviews to collect data for analysis, according to Giorgi (2009). However, the qualitative exploratory research method may better support the desired goal of identifying ways that technology managers can retain female computing talent.

Research Question

As stated in Chapter One, a central research question, at a broad level, helps to guide the research (Creswell, 2014). According to Creswell (2014), a central research question helps researchers explore the intricate factors that surround a phenomenon. The central research question in this study is:

What are the strategies technology managers need to retain women in the computing industry?

Population and Sample

The population in a research study is a group from which we select the sample, based on a defining characteristic (Banerjee & Chaudhury, 2010). Given that the central research question focuses on experiences women in the pursuing a career in the computing industry, the ultimate target population would be the individuals working in that industry. The population of women in computing jobs in San Diego consists of thousands of women. This population was appropriate for two reasons. The first reason being that it was easily accessible to the researcher. The second reason being because it provided an adequate representation of the gender (women) and profession (computing) being studied which, according to Banerjee & Chaudhury (2010), would allow for the selection of an appropriate sample to make inferences about women in computing.

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A sample is a subset of a target population that researchers collect data from (Banerjee & Chaudhury, 2010). Sample size is the number of elements in the sample from which data will be drawn and, according to Morse (2000), there is an inverse relationship between the number of participants and the amount of usable data. Sample size is determined by both the type of research and the researcher and can vary, as long as that number allows for data saturation (Charmaz, 2006). According to Charmaz (2006), saturation is dependent upon numerous elements, not all of which are controllable by the researcher, such as timeline, funding, or the experience of the researcher to know when saturation has been achieved. Crouch and McKenzie (2006) argue that in qualitative research that uses in-depth interviews, saturation can be achieved with a sample size less than 20. Additionally, Englander (2012) argued that sample size is only relative to quantitative research because of its dependence on generalizability. Englander (2012) asserts that because qualitative research does not seek generalizability, that a minimum sample size should not exist; however, it is recommended that at least three participants be used. The

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sample size for the study of 10 is appropriate because Walters (2015) and Clagget (2016) conducted similar studies where a sample size of 15 and 11 were used.

Sampling Procedure

Lofland, Snow, Anderson, and Lofland (2006) suggested that purposeful sampling, over random or probability-based sampling, is the more common type selected for qualitative research because the population size is often unknown or the researcher is looking to identify variation across several cases. Purposeful sampling allows for the researcher to select participants based on specific qualities that support the intended research (Etika, Musa, & Alkassim, 2016) and was appropriate for this study because it specifically targets a demographic based on gender and profession.

After IRB approval, potential participants were contacted using emails or phone numbers obtained from personal and professional contacts working in the region. Participants were characterized by establishing selection criteria that will provide the most relevant or supportive information to the identified research topic (Marshall, 1996). The selection criteria for participants is that they met the identified demographic characteristics. The required demographic characteristics were women who have successfully obtained employment in the computing profession.

Participants were contacted by email or phone requesting their participation. When a participant agreed to participate in the study, the informed consent form (see Appendix A) was sent to the participant. Next, interview times and dates were established with each study participant.

Participants were informed of their right to withdraw from the study at any time. For the final report, participants were referred to only by pseudonyms. Data and records were stored in a locked filing cabinet and on password protected computers located in secure locations, and these

materials were accessible only to the primary researcher. The data will be kept securely stored for two years.

Instrumentation

For qualitative research, the researcher is an implicit part of the research and must collect valid and reliable data, according to Denzin & Lincoln (2003). Denzin & Lincoln also assert that qualitative researchers often serve as the instrument. Qualitative research uses open-ended questions.

During data collection, a digital audio recording device was used to capture data. The interview process used interview questions to capture the interviewee's perceptions (see Appendix B). The interviews lasted between 30 and 90 minutes. The interviews were conducted using the established interview protocol (see Appendix C).

Open-ended, semistructured interview questions were used to explore how strategies might be developed that technology managers could use to retain female computing talent. DiCicco-Bloom & Crabtree (2006) noted open-ended interview questions encourage participants to provide detailed descriptions of their lived experiences. Depending upon the responses provided, follow-up or probing questions were used to capture additional details (DiCicco-Bloom & Crabtree, 2006). One-on-one interviews took place at a date and time convenient to each participant.

Handwritten notes of each participant's interview were captured. These notes were used to identify information for follow-up questions, potential strategies or strategic elements, and even non-verbal cues. Notes, along with transcripts of the interviews were made available to the interviewee for the purpose of member checking.

Labeling of the captured data was used to ensure the proper categorization of themes and ideas that result from participant responses (Renner & Taylor-Powell, 2003). According to

Renner and Taylor-Powell (2003), the labeling or categorization represents the core element of the qualitative research process. They asserted that labeling activities are instrumental in identify themes and organize them into easily definable categories. Labeling activities were conducted using anonymous identifiers (e.g., P1, P2, ..., P10). At no time was personal identifying information captured.

To ensure that participants were not harmed, the research adhered to the components of an ethical of research relationship. According to Hewitt (2007), these components are acknowledgement of bias, rigor, rapport, respect for autonomy, avoidance of exploitation, and confidentiality. Data collection and analysis was conducted with consideration of these elements and how the interpretation and reporting of any data as a result of the research process might adversely affect participants.

Validity

Validity differs depending upon the type of research being conducted. According to Creswell (2014) validity in quantitative research uses statistical tools and measurements to determine validity, while qualitative research uses field research and data collection from individuals to determine common themes. Creswell (2014) pointed out that validity is especially important for a qualitative study because it substantiates that the researcher actually observed what they think that they have observed. Validity was demonstrated by following steps similar to that of Clagget (2016) by having interview questions reviewed by subject matter experts, as well as leveraging audio recordings to ensure that data is accurately collected.

The key to establishing credibility lies in the ability to prove that a study measures what is intended, according to Shenton (2004). DeVault (2017) pointed out that there were additional attributes of credibility that should be present, such as member checks, transcript review, and triangulation. According to DeVault (2017), member checks occurred when participants were

asked to review the collected data from other participants and provide their interpretation. Member checking is different from transcript review, which is the process of providing participants with the verbatim transcript of the interview for them to validate and confirm that the information was collected accurately (Hagens, Dobrow, & Chafe, 2009). Additionally, DeVault (2017) stated that triangulation occurred when the researcher asked the same questions to different participants or by collecting data from various sources, using different methods.

DeVault (2017) stated that reliability is dependent upon validity and that some researchers feel that if that credibility, as defined above, has been achieved, then demonstration of dependability on its own is unnecessary. Shenton (2004), however, stated that it is critical for a researcher to demonstrate that, given the same context, method, and participants, similar results would be achieved. Both DeVault and Shenton pointed out the importance of detailed notes regarding the methods, approaches, and processes used in the collection of data, should the dependability of the research ever be questioned.

DeVault (2017) explained transferability as the application of research findings to other situations and contexts. DeVault (2017) pointed out that the context in which we collect the data is critical to interpretation, which forces qualitative research to be very narrow in scope. Additionally, DeVault (2017) asserted that purposeful sampling was essential to preserving the contextual integrity, which is one reason that purposeful sampling was the sampling method for this research study.

Shenton (2004) described the concept of confirmability regarding objectivity. More specifically, achieving confirmability when the researcher can entirely remove all biases (implicit or otherwise) from the results of the data analysis. Shenton (2004) emphasized the

effectiveness of triangulation in the data collection process to help illustrate that the results are free from researcher bias.

Reliability

Reliability relates to the ability of a measuring instrument to be free of random or unstable error (Emory & Cooper, 1991). Reliability refers to the extent of the consistency of results, in terms of trustworthiness and quality (Golafshani, 2003). DeVault (2017) states that reliability is dependent upon validity and that some researchers feel that if that credibility, as defined, has been achieved, then demonstration of dependability on its own is unnecessary. Shenton (2004), however, stated that it is critical for a researcher to demonstrate that, given the same context, method, and participants, similar results would be achieved. Both DeVault and Shenton pointed out the importance of detailed notes regarding the methods, approaches, and processes used in the collection of data, should the dependability of the research ever be questioned.

Conducting member checking enhances the reliability and validity of the data collection process, according to Shenton (2004), some of which may occur during the interview to ensure accuracy. After reviewing the interview data, the researcher provided a transcript of the data, along with any analysis or information synthesis that occurred as a result of the interviewee's responses, to determine whether or not that analysis accurately reflects the information provided (Creswell, 2014). Researchers use member checking to validate study findings (Shenton, 2004). Walters (2015) used member checking to conduct debriefing sessions with interviewees to allow for the correction of misconceptions or misinterpretations.

Using triangulation contributes to the overall reliability of a study by identifying categories and themes using multiple resources (Shenton, 2004). This process aids in interpreting data reliability because different sources limit their individual weaknesses, while exploiting

their respective strengths (Guba, 1981). For the study, triangulation was accomplished by using multiple sources of corroboration, such as other researchers and theories (Walters, 2015).

Dependability is an aspect of research that supports reliability in the context of being able to reproduce results of the study (Shenton, 2004). According to Shenton, dependability in qualitative research does not always mean that the exact results can be obtained, due to the largely explorative nature of the tradition. Therefore, the study could act as a model or prototype for future researchers to apply towards similar studies. For this study, dependability was addressed by using clearly defined research processes that described planned and executed actions, clearly documented activities in data collection, and provided an evaluation of the effectiveness of the process (Shenton, 2004).

Data Collection

The research question allows the researcher to clarify the purpose of the study and, in turn, heavily influences the means by which data is collected (Agee, 2008). The primary research question for the research was:

What are the strategies technology managers need to retain women in the computing industry? The data collection technique selected to answer the research question was semi-structured interviews. Semi-structured interviews provide reliable data and are used in research where the identified concepts are relatively well understood (Given, 2008). In the study, there was a large amount of literature that discussed specific barriers to entry for women in computing in the U.S. These concepts acted as a baseline to be used to compare the experiences between previous studies and the results of this study.

Phone, email, and social media were used to communicate with potential participants to confirm their willingness to participate and coordinate their availability. Participants received the

researchers email, phone number, and social media contact information. Communication with participants occurred via email, phone, and social media.

The interview pool consisted of ten participants from San Diego, California. As specified in the interview protocol (see Appendix C), participants were asked open-ended questions and potential follow-up questions during the semi-structured interviews. The interviews were conducted at a location selected by the participant. Once participation was confirmed, the exact location, date, and time of each interview was established, based on availability and convenience.

Each interview consisted of the following general process: (a) established rapport with the participant; (b) introduced the study, its purpose, and its constraints; (c) obtained a signed consent agreement form (see Appendix A), (d) used the interview protocol (see Appendix C) to ensure all questions were asked and were in the correct format, (e) used probing techniques of the silent probe, overt encouragement, elaboration, clarification, and reflection; (f) thanked the participant for their time and efforts.

Interviews were recorded using a digital recording device. A personal journal was used during the interview to capture notes and memos. After the interviews were concluded, the recorded information was transcribed to Microsoft Word using the transcription process. This process involved listening to recorded interviews and creating a verbal to written translation for further analysis.

Data Analysis

Data analysis consisted of activities in three general areas: coding, categorizing and memoing. Coding activities were conducted using Thomas' (2006) five-step process for codifying data:

1. Preparation of raw data

2. A close reading of the text
3. Creation of categories
4. Overlapping coding and uncoded text
5. Continuing revision and refinement of category system (p. 241-242)

Due to the open-ended nature of the research question, it is considered exploratory.

Additionally, the coding process followed an epistemological process in which phenomenon was identified and categorized for use in drawing conclusions and developing hypotheses.

Categorization activities included grouping codes based on emergent themes and specific concepts. The frequency of themes or concepts was noted, as well as whether or not a discernable sequence of events was predictable. Lastly, individual codes were assessed for overarching themes.

During the interview process, notes and observations of the process were recorded by this researcher using memos. During the data analysis process, these notes and observations were revisited to provide contextual information for use in the analysis. Additionally, these items were used in the categorizing process to help identify themes and concepts, as well as to indicate sequences.

The coding and analysis process was consistent with Mansourian's (2008) inductive data analysis process, as seen in Figure 4. The analysis process compared the emergent themes and concepts with the existing literature to isolate new findings. It also included additions to current results to generate new data.

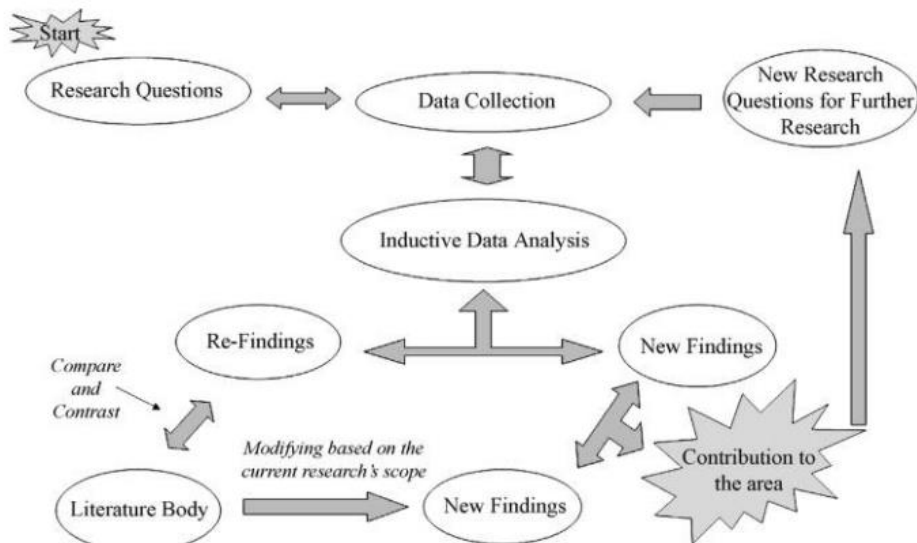


Figure 4. Exploratory Research Data Collection and Analysis. Reprinted from *Exploratory nature of, and uncertainty tolerance in, qualitative research* (p. 279) by Mansourian Y., 2008, *New Library World*, 109, p.279. Copyright 2008 by New Library World. Reprinted with Permission.

Ethical Considerations

The responsibility of a member of the academic research community is to conduct research ethically, with full honesty, respect, and care for the well-being of those who participate in the study. This study was designed to uphold that responsibility. Before beginning recruitment, Institutional Review Board (IRB) approval was obtained. The principles of the *Belmont Report* protocol ensures the highest level of ethical research. The *Belmont Report* principles focus on the well-being of study subjects (Bromley, Mikesell, Jones, & Khodyakov, 2015). Specifically, the principles of justice, respect, and beneficence were apparent in the study.

The vulnerable research population was protected from potential exploitation (Rogers & Lange, 2013). The three principles from the *Belmont Report* are addressed in the study, to ensure ethical practices are maintained.

Justice. The *Belmont Report* associates the principle of justice in research, in the context of exploitation (United States, 1978). According to the report, researchers should not select subjects based on a predisposition of results, availability, and ease of manipulation. The research does not include a population or sample that fits the criteria discussed in the *Belmont Report*; instead, it targeted explicitly women working in the computing industry in the identified geographical region.

Respect. The *Belmont Report* points out two ethical convictions to be concerned with in regards to respect for persons (United States, 1978). According to the report people fall into two categories, those who can think and act for themselves (autonomous agents) and those who do not. The former group is fully capable of contributing to research without exterior pressures or agendas. The latter category need to be protected and were not selected to participate in research because they had the potential to be coerced into providing inaccurate data. The research only included subjects that were considered autonomous.

Beneficence. The *Belmont Report* provides guidance on not just protecting subjects from harm, but also to protect their well-being, similar to the Hippocratic maxim of "do no harm" (United States, 1978). Additionally, the purpose of the research should also be to benefit the greater population of which the subjects are a member. Each participant was required to sign an informed consent form (see Appendix A), to ensure awareness of the risks and benefits of the study. The consent form included (a) the purpose of the study, (b) the involvement of participates, (c) participation procedures, (d) the benefits of the research, (e) the risks of taking

part, (f) costs and compensation, (g) confidentiality, (h) voluntary nature of participating, and (i) the rights of the participant to withdraw (Wright, 2011).

Summary of Chapter Three

The use of a qualitative exploratory method fits well with the identified phenomenon of a decline of women in computing because the number continued its decline, despite the amount of research already conducted on the topic. By focusing on women who have successfully navigated the path of becoming proficient with the skills needed to achieve employment and asking them to share their experiences and ideas on why they were successful, it was possible to identify key strategies that current technology managers can employ to retain new female computing professionals. The next chapter will present the data collected during the research process.

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CHAPTER FOUR

The purpose of this study was to explore the strategies technology managers need to retain women in the computing industry. There are estimated shortages in crucial computing professions, such as cybersecurity, which Kauflin (2017) estimated as having a shortage of two million qualified candidates by 2019. Another growing computing discipline, big data, could face shortages of 140,000 – 190,000 workers with the appropriate skillset, according to Manyika et al. (2011). These shortages, amongst others in the industry, represent well-paying jobs that could potentially be filled by women. While much research has been conducted regarding the issue of gender balance in computing, what remains is a gap in the literature that solicits direct input from women in the computing industry as to how their experience has allowed them to overcome or mitigate the identified barriers to entry, as well as let them remain in the industry successfully. To address this gap and to make a practical contribution to the practice of management, this study posed eleven open-ended questions, with corresponding sub-questions (see Appendix B), based on a semistructured interview approach, to elicit the perspectives and viewpoints of technology managers to explore the question central to this research: What are the strategies technology managers need to retain women in the computing industry?

This chapter presents analysis of the data collected from the exploratory study using a semi-structured interview method. First, the chapter outlines the demographics of the study participants. Second, discussion of the qualitative data collection process and how that data were analyzed will occur. The presentation and discussion of the data align with the interview questions and sub-questions. Finally, this chapter summarizes the analyzed data that will support all study finding, conclusions, and recommendations for future research that exist in Chapter Five of this dissertation.

Participant Demographics

Table 1

Study Participant Demographics

ID	Title	Career Level	Company Size	Origin
P1	Data Scientist	Mid	Large	USA
P2	Mgr, Electrical Engineering eCAD	Senior	Large	USA
P3	Certified IPC Designer	Mid	Small	USA
P4	Mgr, Software Development	Senior	Large	Italy
P5	Network Engineer	Mid	Medium	USA
P6	Software Development Engineer	Junior	Large	USA
P7	Systems Engineer	Mid	Medium	USA
P8	Telecommunications Specialist	Senior	Medium	Georgia
P9	Senior Software Engineer	Mid	Large	China
P10	Software Engineer	Mid	Medium	Iran

Note. eCAD = Electrical Computer Aided Design; IPC = Institute for Printed Circuits;
 Junior: < 5 years; Mid: 5-15 years; Senior: > 15 years;
 Small: < 50 employees; Medium 50 -500 Employees; Large > 500 Employees
 Origin = Country of Birth.

The study participants consisted of 10 women who were purposively recruited, selected, and interviewed for this study. To maintain anonymity and confidentiality, each study participant was identified using code names (P1, P2,..., P10). Although participant country of origin was not addressed in this study, nor was it the intent of this study, the sample population contained four out of ten women who were born and received education and training in a country other than the

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United States. The interviews were conducted in the Greater San Diego, California region between the dates of November 14th, 2018 and April 5th, 2019.

The official titles of the women who participated in this exploratory qualitative study were: Data Scientist, Electrical Engineering eCAD Manager, Certified IPC Designer, Software Development Manager, Network Engineer, Software Development Engineer, Systems Engineer, Telecommunications Specialist, Software Engineer, and Senior Software Engineer. The participants had varied experience levels. One participant out of ten was considered junior level with less than five years of professional computing experience. Six participants out of ten were considered mid-level with between five and fifteen years of professional computing experience. Two participants out of ten were considered senior level with greater than 15 years of professional computing experience. Additionally, the participants were employed by companies of varied size. One participant out of ten worked at a small company of fewer than 50 employees. Four participants out of ten worked for medium-sized companies that employed between 50 and 500 personnel. Five participants out of ten worked for large-sized companies that employed greater than 500 employees.

Presentation of the Data

This section presents the qualitative data collected during participant interviews with ten women who met the criteria for acceptable representations of the target population and sample. This study posed eleven open-ended questions, each with one to three sub-questions (see Appendix B), using a semi-structured interview approach. The approach was chosen to evoke the perspectives, viewpoints, and experiences of women working in the computing industry to explore the central research question in this study: What are the strategies technology managers need to retain women in the computing industry? Data saturation was reached upon completion

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of the 10 participant interviews. Each interview was conducted at a time and location agreed upon by the participant and researcher. The interviews were recorded using a digital recording device and field notes were recorded by the researcher to support triangulation. Upon completion, each interview was transcribed, with member checking and study participant transcript review occurring to validate study participant response and to ensure accuracy and credibility.

Taylor-Powell and Renner (2003) asserted that grouping participant responses by the questions or topics of the research could help to focus the qualitative data analysis. According to Taylor-Powell and Renner (2003), the process of grouping was especially helpful when dealing with the large amounts of rich, qualitative data that open-ended questions yield. This study used grouping in an effort to focus the analysis of the rich qualitative data that was collected during the the participant interviews, allowing for the exploration of the central research question. The ten open-ended questions were grouped into five subgroups: (a) computing background, (b) experiences with gender balance, (c) identity alignment and self-efficacy, (d) management and mentorship, and (e) positive and negative organizational factors. Each subgroup represents a topic that was explored in the literature review and relates to the conceptual framework.

Stuckey (2015) argues that the coding process can be both inductive and deductive. In vivo codes could evolve from analysis of the repetition of words, phrases, or concepts in the data, while a priori codes could be developed from foundational research or a conceptual framework. Data analysis was conducted using both inductive (a priori) and deductive (in vivo) coding.

Subgroup 1: Computing Background

The topic of computing background was explored using one primary question, with two follow-up questions:

1. Describe what led you to pursue a career in computing?

- a. Were you encouraged to pursue this career?
- b. What interests you most about computing?

Data aggregated for the topic of computing background was grouped into three themes:

(a) Career Interest, (b) Career Encouragement, and (c) Career Deterrence. The number of participant responses associated with the themes for this topic is presented in Table 2.

Table 2

Themes for Computing Background

Subgroup themes	Number of participants who discussed the theme
Career Interest	10/10
Career Encouragement	10/10
Career Deterrence	4/10

Subgroup 1, Theme 1 (Career Interest). An interest in a career in computing was a prevalent theme in the data analysis of the computing background, with all ten participants discussing the theme. Two categories developed under the theme of Career Interest: (a) problem solving and (b) emerging technologies. Attraction to the problem solving aspect was discussed by five out of the ten participants, while interest in emerging technologies was identified as a career attractor by two out of the ten participants. Excerpts from the transcribed study participant interviews discussing Career Interest are presented in Appendix D.

Subgroup 1, Theme 2 (Career Encouragement). As with Career Interest, Career Encouragement was a prevalent theme, appearing in all ten participant interviews. Two

categories developed under the theme of Career Encouragement: (a) familial encouragement and (b) teacher encouragement. Familial Encouragement was the most common category discussed by four out of ten participants. Self-encouragement was the second most common category identified by participants, with four out of ten identifying this method. Excerpts from the transcribed study participant interviews discussing Career Encouragement are presented in Appendix D.

Subgroup 1, Theme 3 (Career Deterrence). Career Deterrence was a less discussed theme with the study participants. Four out of ten participants provided examples of Career Deterrence. It is important to note that this theme and the theme of Career Encouragement were not mutually exclusive, as the same four participants also cited examples of Career Encouragement. Two categories developed under the theme of Career Deterrence: (a) educational deterrence and (b) workplace deterrence. In both categories, two out of ten participants provided examples of behaviors that adversely affected their journey in gaining employment in the computing industry. Excerpts from the transcribed study participant interviews discussing Career Deterrence are presented in Appendix D.

Subgroup 2: Experiences with Gender Balance

The topic of experiences with gender balance was explored using three primary questions, with four follow-up questions:

1. Describe your background with computing training/education.
 - a. How would you describe the gender balance in your training/education?
2. In your current organization, how would you describe the gender balance?
 - a. How does this compare to previous organizations that you have worked for?
3. What is your awareness regarding the gender balance in computing?

- a. Why do you feel that the number of women in computing is declining?
- b. Do you feel like you will remain in the computing industry?
 - i. If yes, why will you remain?
 - ii. If no, what will cause you to leave?

Data aggregated for the topic of experiences with gender balance was grouped into three themes: (a) Gender Balance in Education and Training, (b) Gender Balance in the Workforce, and (c) Female Retention in Computing. The number of participant responses associated with the themes for this topic is presented in Table 3.

Table 3

Themes for Experiences with Gender Balance

Subgroup themes	Number of participants who discussed the theme
Gender Balance in Education and Training	10/10
Gender Balance in the Workplace	10/10
Female Retention in Computing	9/10

Subgroup 2, Theme 1 (Gender Balance in Education and Training). Every participant in this study acknowledge completing some form of higher education or computing-specific training. Table 4 presents short descriptions used by participants when describing the experiences with Gender Balance in Education and Training. The majority of participants expressed an overwhelming majority of male students. However, three of the participants that received formal education outside of the United States (Iran, Italy, and Georgia) reported a closer

ratio of male to female students. Excerpts from the transcribed study participant interviews discussing Gender Balance in Education and Training are presented in Appendix D.

Table 4

Study Participant Descriptions for Gender Balance in Education and Training

Short descriptions	
The majority were males	Four in a class of twenty-five
I'm normally the only girl	10 to 15% women
Probably 40% of women	Sometimes there was one other girl
I was usually the only girl	One of two girls in one hundred guys
Very Imbalanced	Maybe two women out of ten
It was more balanced in Italy	Me and [another girl] out of 30 to 50 students

Subgroup 2, Theme 2 (Gender Balance in the Workplace). Participant responses on the topic of Gender Balance in the Workplace were similar to those given when asked about Gender Balance in Education and Training. Table 5 presents short descriptions used by participants when describing their experiences with Gender Balance in the Workplace. Excerpts from the transcribed study participant interviews discussing Gender Balance in the Workplace are presented in Appendix D.

Table 5

Study Participant Descriptions for Gender Balance in the Workplace

Short descriptions	
It's still a male dominated industry	Definitely an imbalanced work situation
We maybe have 30 women working	There are 35 people my organization

in this office	and there are five females
I rarely see female technical staff	1% maybe less
There's only me	I would say that there's no balance
Maybe 2%	It's probably about 5 to 10%
Still pretty imbalanced. Probably I'd	There are five of us and I'm the only female
say there is 20% women	

Subgroup 2, Theme 3 (Female Retention in Computing). The last theme discussed in this subgroup is that of Female Retention in Computing. When presented with a question asking participants why they thought women did not persist in the computing industry, five categories emerged: (a) Glass Ceiling/Lack of Female Role Models, (b) Bias, (c) Identity Alignment, (d) Self-efficacy, and (e) the Desire to Have a Family. The developed categories were not mutually exclusive, with some participants providing responses that fit into more than category. The number of participant responses associated with the categories for this theme is presented in Table 6. Excerpts from the transcribed study participant interviews discussing Female Retention in Computing are presented in Appendix D.

Table 6

Study Participant Responses for Female Retention in Computing

Category	Number of participants who discussed the category
Glass Ceiling/Lack of Female Role Models	4/10
Bias	4/10
Identity Alignment	3/10
Self-efficacy	3/10
Desire to Have a Family	2/10

Subgroup 3: Identity Alignment and Self-efficacy

The topic of identity alignment and self-efficacy was explored using one primary question, with three follow-up questions:

- 1. How would you describe a typical person working in the computing industry?
 - a. How do you compare to this description?
 - b. Do you feel like you fit in with your peers in the industry?
 - c. How proficient do you feel in comparison with your peers?
 - i. If no, what will cause you to leave?

Data aggregated for the topic of experiences with gender balance was grouped into three themes: (a) Described Stereotype, (b) Identity Alignment, and (c) Self-Efficacy. All but one participant responded to these questions, with that one participant preferring not to respond to the question regarding a stereotypical computing worker.

Subgroup 3, Theme 1 (Described Stereotype). When asked to describe a stereotypical person working in the computing industry, participants responded with strikingly similar descriptions. The number of participant responses associated with the description for this theme is presented in Table 7. Excerpts from the transcribed study participant interviews discussing Described Stereotype are presented in Appendix D.

Table 7

Study Participant Descriptors for Stereotypical Computing Worker

Theme Descriptors	Number of participants who used the descriptor
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Male	7/10
Smart or Geeky	6/10
Not Social	5/10
Passionate About Computing	5/10
White	3/10
Asian	1/10

Subgroup 3, Theme 2 (Identity Alignment). After providing a personal description of a stereotypical person who worked in computing, participants were asked how they felt that they compared to that description. Half of the women responded that they felt like they didn't fit their described stereotype. The other half did feel that they compared closely, with the exception of their gender. One participant declined to provide a stereotype description and therefore could not provide an answer to this question. All of the participants responded that they felt that they fit in with their peers. One participant stated that it took some time to feel that she fit in. One participant stated that she felt it was easier for men in the industry to bond. Excerpts from the transcribed study participant interviews discussing Described Stereotype are presented in Appendix D.

Subgroup 3, Theme 3 (Self-Efficacy). Participant responses on the topic of self-efficacy, or how proficient they were in their job were given in context of how well they compared themselves to their peers. Table 8 presents short descriptions used by participants when describing their self-efficacy in comparison to their peers. Excerpts from the transcribed study participant interviews discussing Self-efficacy are presented in Appendix D.

Table 8

Study Participant Descriptions for Self-efficacy

I think I'm pretty even.

I think that I may be in the middle where I'm still learning, but I also know a lot.

Not very anymore only because this company... is in a different type of industry.

I was more proficient than most of the people.

I'm usually the go to person to ask how to solve problems.

I'm probably one of the best Network Engineers.

Like maybe 75 to 80%.

I'm pretty proficient.

In many cases I have to transfer knowledge to my teammates who are much less proficient than I am.

I don't feel I'm like the top dog, ... but I don't feel I'm the bottom either. Toward the top.

I think I'm pretty proficient.

Subgroup 4: Experiences with Management and Mentorship

The topic of experiences with management and mentorship was explored using two primary questions, with three follow-up questions:

1. Do you prefer to have a manager that is the same or different gender?
 - a. What factors determine this preference?
2. Describe your experiences with professional mentorship.
 - a. What experiences do you have being a mentee?
 - b. What experiences do you have being a mentor?

Data aggregated for the topic of experiences with gender balance was grouped into two themes:

(a) Manager Gender Experiences and Preferences and (b) Mentorship Experiences and Gender Preferences. The latter theme covers both experiences as a mentor and a mentee.

Subgroup 4, Theme 1 (Manager Gender Experiences and Preference). The purpose of this thematic group was twofold. Firstly, it determined participant experience with both male

and female managers. The second purpose was to determine if participants had a preference for either gender in a management role. Table 9 presents participants manager gender experiences, as well as their preference for either a male or female manager. Excerpts from the transcribed study participant interviews discussing Manager Gender Experiences and Preference are presented in Appendix D.

Table 9

Study Participant Manager Gender Experience and Preference

Study			
Participant	Male Manager	Female Manager	Manger Preference
P1	YES	YES	MALE
P2	YES	YES	NONE
P3	YES	YES	MALE
P4	YES	YES	NONE
P5	YES	YES	NONE
P6	YES	YES	FEMALE
P7	YES	YES	NONE
P8	YES	NO	NONE
P9	YES	NO	NONE
P10	YES	NO	NONE

Subgroup 4, Theme 2 (Mentorship Experience). All study participants were familiar with the concept of mentorship and either had experience with being a mentor or mentee, or had some interest in formal mentorship. Table 10 presents participant experience with Mentorship. Excerpts from the transcribed study participant interviews discussing Mentorship Experiences are presented in Appendix D.

Table 10

Study Participant Experience with Mentorship

Mentorship Experience	Number of participants with Experience
Mentee Experience with a Male Mentor	8/10
Mentee Experience with a Female Mentor	3/10
Mentor Experience with a Male Mentee	6/10
Mentor Experience with a Female Mentee	6/10
No Experience with Mentorship, but Interested	2/10

Subgroup 5: Positive and Negative Organizational Factors

The topic of positive and negative organizational factors was explored using two primary questions, with four follow-up questions:

1. What, if any, strategies or programs have been implemented by your current or previous managers that have contributed to your success in the computing industry?
 - a. How specifically have they helped you to succeed?
 - b. How could they be altered to further support your success?
2. What, if any, strategies or programs have been implemented by your current or previous managers that have limited your success in the computing industry?
 - a. How specifically have they limited your success?
 - b. How could they be altered to further support your success?

The two themes discussed in the Positive and Negative Organizational Factors subgroup are: (a) supportive strategies and (b) limiting factors. The number of participant responses associated with the themes for this topic is presented in Table 11.

Table 11

Themes for Experiences with Positive and Negative Organizational Factors

Subgroup themes	Number of participants who discussed the theme
Supportive Strategies	8/10
Gender-Specific Supportive Strategies	5/10
Limiting Factors	6/10
Gender-specific Limiting Factors	3/10

Subgroup 5, Theme 1 (Supportive Strategies). A majority of the participants responded that they had been exposed to an organizational strategy that they found supportive to their career, with seven out of ten participants identifying some sort of supportive strategy. Five out of ten participants identified gender-specific strategy, in the form of a women's group, that they felt was beneficial to their success. Excerpts from the transcribed study participant interviews discussing Supportive Strategies are presented in Appendix D.

Subgroup 5, Theme 2 (Limiting Factors). None of the participants felt that there were organizational strategies that limited their success. However, participants did provide examples of factors that they felt limited their success. Six out of ten participants felt that there were organizational factors that limited their success. Three out of ten participants provided examples of gender-specific limiting factors. Excerpts from the transcribed study participant interviews discussing Limiting Factors are presented in Appendix D.

General Comments

At the end of each interview, the participant was asked the following question:
Is there anything else that you would like to add regarding your experiences with:

1. Computing education
2. Obtaining a job in computing
3. Remaining in the computing industry

Several participants (P1, P4, P6) suggested implementing outreach programs to schools, starting at the elementary level, to help foster and support interest in computing for young girls and women. Another (P6) pointed out the importance for women to set specific long-term goals, instead of general long-term goals early in career development, as failing to do so could be hurtful to career development. She also suggested that job requirements should be reviewed by a diverse set of people (for example multiple genders or cultures) to ensure that those requirements don't alienate potential candidates. This echoes sentiments provided by another participant (P9), who states that she wouldn't apply to certain jobs if she felt that she didn't meet most, if not all of the requirements. Likewise, one participant (P5) suggested programs designed to address self-efficacy concerns when recruiting women. Two participants (P1, P6) suggested improving maternity leave programs that would prevent women from having to choose between a career and a family. Last, several participants (P3, P4, P6, P8) pointed the need to create programs designed to eliminate the feeling of intimidation that women face, whether from peers, work environment, or job challenges. Excerpts from the transcribed study participant interviews discussing general comments are presented in Appendix D.

Presentation and Discussion of Findings

This study explored the central research question: What are the strategies technology managers need to retain women in the computing industry? Beyer (2014) stated that despite much research into the phenomenon of declining numbers of women in computing, there has yet to be a successful strategy to help deter this decline. This study looks at the phenomenon through

the lens of Social Identity Theory (SIT), developed by Tajfel and Turner (1979), specifically linking the concepts of identity alignment and self-efficacy to SIT's in-group/outgroup mindset. This study also used Wacjman's (2000) criticism of lack of female representation in the computing industry as a reflection of the implicit (or in some cases, explicit) gender bias. That bias, when coupled with Pinch and Bijker's (1984) social construction of technology (SCOT) theory, illuminates an industry that could be viewed anywhere from unattractive to outright hostile.

A hybrid approach was used to analyze the data. Elements found in the conceptual framework presented in Figure 3 were developed into interview questions (See appendix B), which resulted in the a priori subgroups: (a) computing background, (b) experiences with gender balance, (c) identity alignment and self-efficacy, (d) management and mentorship, and (e) positive and negative organizational factors. Several follow-on questions are also represented by a priori themes, with other in vivo thematical groupings emerging from analysis of the participant responses. Field notes and jottings made by the researcher during the interview process were added to analyzed data to create the rich qualitative data needed to support the study. Discussion of the findings will follow the same groupings as presented in the previous section.

Computing Background

While the focus of this study was to develop strategies for technology managers to help retain women in the computing industry, it would be remiss to not review those factors that led those women to seek and obtain a career in the industry. Especially given the few number of women who overcome the barriers discussed in the literature. The most common element amongst the participant responses was attraction to the problem solving nature of jobs within industry. For example, one participant stated, "When I got more into coding (I code in R and

Python) it was its own kind of level up to puzzle out ... that's kind of what appeals to me is that it's more of a game rather than like hard work" (P1).

Other themes prevalent in the data were that of career encouragement and deterrence. Participants indicated that familial and teacher encouragement was common. For example, one participant stated, "My school teacher was thinking that I would be good in math in general and I was also encouraged by my parents" (P7). However, she went on to add, "I can't say that my parents were the ones who made this decision for me. I was also on the same page." This latter statement gives credence to the importance of self-encouragement which was a common theme that was discussed by participants when asked about career deterrence. For example, one participant stated "when I first started you were the only woman in the group. You are asked to go make coffee, you are asked to go make blueprints. Dirty jokes abounded that were very inappropriate and you didn't have anyone to go to, you just had to stick it out, you know, and I think that's one reason there weren't many women who could stick that out and make it through" (P2).

Experiences with Gender Balance

The second subgrouping explored the gender balance experiences of the participants during formal education and training environments, such as college, trade school, or certification courses. It also looked at the gender balance in current and previous places of employment. In both cases, the gender balance reflected the overabundance of men that has been identified in the literature. Participant responses regarding the imbalance were similar, with short descriptions such as, "I was always the only woman (P1)", "Mostly men (P3)", and "The majority were males (P9)." Four participants received education outside of the United States and reflected that the gender balance was more even in those countries. One participant stated, "When I was in Italy it was more balanced ... there was a good ratio between men and women (P4)." Another

participant stated, “Back in college [in the country of Georgia] it was like probably 40% of women 60% of men (P7).”

Participant responses for experiences with gender balance in the workplace aligned with both their own educational experiences and supported findings in the literature. Example participant responses were, “I would say there are ... 5 to 15% of that building as women, out of everybody there (P1)” and “I would say I would say there's no balance really in my personal experience ... most job sites that we're installing a job or training a customer or installing an application, I rarely see female technical staff (P8).”

Additionally, participant responses indicated concerns with identity alignment and self-efficacy which were both elements identified in the literature. Examples of this were “I think it depends also on like what industry, like video games are very much male dominated. So it's very different. You almost need to be a tomboy to fit in (P1)” and “I would assume that maybe [women] don't think they can. Maybe they think that it's a little too much or maybe people are just technically challenged or are afraid of the computer. I know so many people that are just afraid of it for some reason (P8).” The emergence of identity alignment and self-efficacy was interesting as they have their own subtheme, which will expand more upon the influence of these concepts on women in computing.

The last aspect of gender balance experiences focused on factors that participants felt influenced their own or other women's decision to remain in the computing industry. Participant responses reflected those found in the literature. For example, one participant noted that the lack of role models was discouraging when she stated, “I don't feel like it's a place for me. Like I don't fit there. I don't belong there and then there is the problem that there are not a lot of role models (P4).” Other participant responses supported widespread bias, such as “For my first job

that I got in IT, the network engineer there flat out told me that he didn't believe that women belonged in IT and that I should get out now. My first job! And that was really hard (P5)."

Identity Alignment and Self-efficacy

To help establish a baseline against which to measure identity alignment, participants were asked to provide their description of a stereotypical person working in the computing industry. Responses were overwhelmingly supportive of the asocial, singularly focused, white, male that is prevalent in the literature. One participant described, "Geeky? Typically, not very social. Detail oriented. And focused I guess. A man (P2)." Another presented, "I mean I see a man right. I work with mostly men. I don't know ... someone who's like really always on their computer. That stereotype of the nerd who sits in the basement and does nothing ... that nerd who sits in the dark and is on their computer (P6)."

With their own description in mind, the participants were then asked to compare themselves to that stereotype. Participant responses indicate that they all felt that they aligned well with their description, with two major exceptions. The obvious first difference being their gender. The second difference was that they felt more outgoing or social than their stereotype. For example, one participant stated, "I'm the opposite and I was even told through the years, 'It's weird like you, you know, you're too out going to be a programmer' (P4)." Another participant noted, "Well, I'm pretty much same other than I'm a woman (P7)."

The last section of this subtheme focused on the issue of poor self-efficacy ratings, which was an issue prevalent in the literature. Participants were asked to compare their proficiency in computing to their peers. The responses overwhelmingly showed that the participants felt that they performed equally or even better than their peers. One participant stated, "I think I'm pretty I'm pretty even (P1)." Another offered, "I think they've been doing this a lot longer than I have. So I think that I may be like in the middle where I'm still learning, but I also know a lot (P3)." A

final participant stated, “I don't like to praise myself, but I'm pretty proficient. In many cases I have to transfer knowledge to my teammates who are much less proficient than I am (P7).” These responses are not unexpectedly contrasting the self-efficacy problem identified in the literature. Having successfully avoided or overcoming self-efficacy issues, all study participants have gained employment in the industry and have stated their intention to remain in the industry.

Management and Mentorship

Extending the concept of identity alignment and self-efficacy, the next subtheme investigated the influence of gender in the realms of management and mentorship, asking the participants to share their experiences with managers, mentors, and mentees. Every participant had experience with male managers. All but three participants identified as having had a female manager at one time. When asked for a preference, only one participant had a preference for a female manager, stating, “I've actually had a really good experience having a female manager. I think I can attribute part of that to her personality in the sense that she's incredibly empathetic and it was very, very easy to connect with her on a personal level and also an emotional level (P6).” Two Participants stated a preference for a male manager, saying “Different 100% (P1) and “I think I would rather have a male supervisor or boss. I guess I get along better with men than I do with women (P3).” The majority of participants had no gender preference for their manager.

Participant experience with mentors (acting as a mentee) echoed their experience with managers, with the majority of participants having experience with male mentors. This is unsurprising given the low number of women in the industry. Only three participants had experience with a female mentor. Two participants had no formal mentorship experience.

Participant experience with mentees (acting as mentor) was more balanced. Five of the participants responded as having experience with male mentees. Six of the participants responded as having experience with female mentees. Only one participant expressed a

preference for mentoring women over men, stating “I really enjoy mentoring people but I choose to mentor women (P1).”

Positive and Negative Organizational Factors

The last subtheme examined in this study focused on organizational strategies, programs, or other factors that participants felt positively or negatively impacted their success in gaining or maintaining a job in the computing industry. While participants were able to provide examples of organizational strategies that they felt supported their career development, there were not examples of implemented organizational strategies that specifically limited their success. They did, however provide examples of limiting factors, which will be discussed.

The majority of participants experienced supportive organizational strategies. One participant stated, “For me, I would say, programs that have helped pay for certifications. For me, getting the certs has certainly helped me be able to master some of the field that I wanted to master (P5).” Additionally, half of the participants mentioned gender-specific strategies that they felt supported them as women in computing. More specifically, participants responded favorably regarding organizational women’s groups. For example, one participant stated, “We have a woman's organization here called the [Company] Women Engineers organization ... What's been really great about it is that it’s actually put me in contact with a lot more women in this office who I don't see on a day-to-day basis (P6).” This was similar to the experience of another participant who said, “I started working with them and their main goal is diversity. We work with a lot of the employee resource groups ... my manager supports it and everybody does and they'll allow me to be passionate about women in technology (P1).”

Conversely, more than half of the participants reported organizational factors that limited their success. For example, one participant stated, “They used to give most of the training to two people. The top two people and then everybody else would get nothing (P8). Several participants

identified that they experienced gender isolation that they felt limited their success. One participant stated, “What we're fighting for in a sense is that it's not that we want women to be necessarily recognized or listened to or whatever. It's just to be treated exactly the same. If you're not listening to the guys, that's fine. If you don't listen to me [because I'm female], like I get it you just kind of an [expletive] (P1).” Another participant stated, “We had to do mandatory sexual harassment training. During the entire training, they singled me out for it. And were like “this is what you can't say. This is what you can't do. This is what happened to [P5].” So I sat there for the entire training basically be different than everybody else (P5).”

Summary of Chapter Four

The results of this study were presented in an effort to answer the central research question: What are the strategies technology managers need to retain women in the computing industry? Data and findings in this chapter were grouped into deliberate subgroups: (a) computing background, (b) experiences with gender balance, (c) identity alignment and self-efficacy, (d) management and mentorship, and (e) positive and negative organizational. These subgroups were developed based key elements found in the literature and extended the ideas contained in the conceptual framework. The grouping process helped to focus the large amount of rich qualitative data collected during the participant interviews. The interview process consisted of a series of semi-structured interview questions, each with one or more follow-up questions. Data collected was compiled and analyzed. A hybrid approach of a priori and in vivo coding was used during the analysis. A priori codes were created from key concepts in the literature and conceptual framework, while in vivo codes emerged from phrases and concepts from participant responses. This chapter presented the resulting data, along with findings derived from that data.

The purpose of this study was to leverage the experience of women who have successfully navigated the process of receiving education and training, as well as gaining employment in the computing industry. While there is no shortage of literature discussing gender balance in the computing industry, many of them focus on STEM as a broader category and fail to look specifically at women in computing in an exploratory, qualitative manner. The data and findings in this chapter are meant to aid technology managers in creating or augmenting strategies designed to help retain the female employees that they do have, and potentially, attract more women to the industry.

However, this study did confirm that while there are still barriers that may prevent women from seeking jobs in computing and remaining in those jobs, those barriers are not insurmountable. Key barriers such as gender bias, identity alignment, and self-efficacy can be overcome by instituting organizational strategies designed to support all genders.

Chapter Five presents a summary of the findings and conclusions of this study, as well as further interpretation and discussion of the results. Additionally, Chapter Five discusses the limitations of the study and further presents the implications for technology managers. Lastly, Chapter Five presents recommendations for actions, applications to professional practice, and future research.

CHAPTER FIVE

The problem addressed in this study is that technology managers lack the strategies needed to retain women in the computing industry (Beyer, 2014). This is evident due to the continued decline of women in the industry, falling to an all-time low, comprising only 25.5% of the 4.8 million computing professionals according to the Bureau of Labor statistics (2017). The shortage of qualified female candidates hinders technology managers' ability to realize the benefits of a diverse team, such as improved innovation and financial results (Nelson, 20014). Lastly, there is a genuine desire for a gender diverse workforce in the United States, with 78% of adults identifying the importance of diversity (Funk and Parker, 2018b).

The purpose of this study was to explore the strategies that technology managers need to retain women in the computing industry. This study examined the experiences of the participants to both evaluate strategies that they had encountered and to develop new strategies that could be implemented to help retain women in the industry. The retention of qualified female technology professionals is key to helping offset shortages in critical computing professions, such as cybersecurity and big data (Kauflin, 2017; Manyika et al., 2011).

This study used a qualitative exploratory approach to answer the central research question and help address a literature gap not currently being addressed by academic or professional literature. Exploratory methods were used to examine the experiences of women in the San Diego, CA technology industry, as well as elicit their viewpoints on the barriers of entry into the industry and their daily struggles with being a woman in a male dominated industry. In contrast to other research, this study focused on a more diverse experience range of female computing professionals, including more women at the entry and mid-career level. Similar studies have been conducted on the much broader STEM grouping (Walters, 2015), or at the managerial or executive level (Clagget, 2016). This study used one central research question, leveraging a

semistructured interview method with eleven open-ended questions, to elicit the experiences and viewpoints of women working in the computing industry in San Diego. The sample size for this study was 10 participants.

This study had several delimitations and limitations. The delimiters for this study were women working in the IT computing industry in San Diego. The first limitation of this study was the concern that participants would not answer questions honestly, instead providing answers that they felt were “correct” or expected answers. This limitation was overcome by explaining the purpose of the study and informing the participants that by being honest and open with their experiences and viewpoints, that strategies could be developed to help future women overcome or even avoid similar barriers to those that the participant faced. The second limitation was that the participants would not feel comfortable sharing their stories with a member of the opposite gender. This limitation was overcome by providing a brief background of myself, my experience in the industry, and my goal in conducting the study when soliciting their participation in the study. Additionally, several of the participants were peers or otherwise associated with each other and were very supportive in sharing their comfort level with the interview process.

As a member of the academic research community, there is a responsibility to conduct research ethically, with full honesty, respect, and care for the well-being of those who participate in the research. Vulnerable research populations must be protected from potential exploitation (Rogers & Lange, 2013). To ensure that exploitation did not occur, and the well-being of participants was upheld, this study adhered to principles set forth in the *Belmont Report*. Three key principles set forth in the *Belmont Report* are justice, respect, and beneficence (United States, 1978). Justice, in the form of ensuring that participants are not selected based on the ability to manipulate responses or outcomes; respect, in ensuring that participants are fully

capable of thinking and acting for themselves and not susceptible to exterior pressures; and beneficence, in ensuring no harm, physical, emotional, or professional, would come to the study participants based on their involvement in the study (United States, 1978).

This chapter presents the findings and conclusions of the conducted study. Next, the chapter further discusses the identified limitations of the study and their impact, if any, on the study results. It also presents the unforeseen limitations that emerged during the execution of the study. Next, this chapter discusses the implications of the study results for industry practitioners and future researchers. Lastly, this chapter summarizes study findings and conclusions.

Findings and Conclusions

This qualitative study explored the experiences of women working in the computing industry to identify strategies that technology managers could use to help retain women in the industry. This study proposed that despite the large amount of literature surrounding the issue of gender balance in computing, there is still a decline of women in the industry, indicating that current technology managers lack effective strategies in retaining (and potentially attracting) women to the industry. By focusing on women who have successfully overcome the barriers to entry for obtaining education and training, as well as successfully obtained a career in the computing industry, it was possible to identify existing, successful strategies and propose new strategies based on the provided experiences. The findings of this study were based on the hybrid analysis of qualitative data collected during the ten study participant interviews and presented several noteworthy insights.

This exploratory study addressed the question: What are the strategies technology managers need to retain women in the computing industry? Eleven open-ended questions, along with several follow up questions, based on a semistructured interview approach, were used to elicit experiences, viewpoints, and perspectives on being a woman seeking, gaining, and

remaining in the computing industry to further identity and develop strategies for current and future technology managers. The questions were divided into subgroups, based on review of the existing literature on women in computing and the conceptual framework presented in Chapter Two, to help guide the analysis of the collected data. This section further expands upon the findings presented in Chapter Four, and presents strategies identified by participants or derived from analysis of the data.

Identified Strategies

This study explored the experiences of women who have successfully obtain education and training, we well as employment in the computing industry. These experiences helped to identify and develop strategies that current and future technology managers can use to retain (and potentially attract) women to the industry. The strategies were identified or emerged from data analyzed in five main categories: (a) computing background, (b) experiences with gender balance, (c) identity alignment and self-efficacy, (d) management and mentorship, and (e) positive and negative organizational factors.

It is important to note that the analysis of the data was contextualized using Tajfel & Turner's (1979) social identity theory (SIT). It also used Pinch and Bijker's (1984) social construction of technology theory (SCOT), more specifically that because women were largely unrepresented in the historical development of technology, they struggle to identify with the technology industry (Wacjman, 2000). The resultant strategies identified and developed from the analysis of data from this study are presented and discussed for each of the five categories.

Computing background. It was important to investigate the computing background of the participants because it provided two key pieces of data. First, it included what motivated the participants to pursue a career in computing. Motivation was key in isolating what attracted women to the industry, and in the context of SIT, what allowed the participants to identify

computing professional as one of the in-groups that they were part of. Second, the elements of encouragement and deterrence were observed to help understand what barriers the participants encountered and how they overcame those barriers to help develop strategies that current technology managers could use to tap into or revive the enthusiasm that lead women to seek a job in computing in the first place.

According to the literature, there are several key issues that prevent women from seeking a career in computing. Perceived stereotype alignment (in-group membership), or the lack thereof was the largest influencing factor on a women seeking computing education (Cundiff et al., 2013; Govender & Khumalo, 2014). Additionally, self-efficacy, or one's own ability to succeed in a specific situation has a tremendous impact on their desire to pursue a career in computing (Beyer, 2014). Familial, teacher, and other mentor support is critical in developing self-efficacy in young women (Heilbrenner, 2013). Table 12 presents potential strategies that a technology manager might implement to engage female employees in the workplace, based on study participant computing background experiences. Table 12 also identifies which participant(s) discussed the theme.

Table 12

Retention Strategies: Computing Background

Theme	Strategy
Computing Background	1. Highlight the problem solving nature of the job (P1, P3, P4, P6, P9, P10) 2. Reinforce the need for female role models in the industry (P1, P2, P4, P10) 3. Implement current or emerging technologies (P2, P7, P10)

Experiences with gender balance. The second category investigated participant experiences with gender balance. Keeping in line with the conceptual model derived from the literature review, analysis looked at gender experiences while obtaining education and training, as well as those experiences that occurred once the participant found employment in the computing industry. The analysis also looked at participant reflections on why the number of women within the industry was so low.

Participant experiences with gender balance in education reflect those discussed in the literature, as well as the data provided by the Bureau of Labor Statistics. Participant responses showed that they were overwhelmingly marginalized both in formalized education, as well as training courses. Of particular note, participant experiences supported existing research regarding the positive effect of exposing women to fundamental computing concepts on increasing interest in Computer Science (CS) classes or a degree in CS (Berg, Apel, & Leicht-Scholten, 2014; Cohoon, Cohoon, & Soffa, 2003; Redmonds, Evans, & Sahami, 2013). Several of the participants noted that they were exposed to computing by their parents, in secondary school, or even early in their undergraduate studies, which supported their continued studies or precipitated a transfer from another degree concentration.

Participant experiences with gender balance in the workplace also mirrored literature and statistical norms. Based on participant responses, women were clearly the gender minority. Their experiences also reflected elements presented in the literature that represent barriers to women in the computing workplace, such as lessened status based on their gender (Cundiff and Vescio, 2016; Karpowitz, Mendelberg, & Shaker, 2012). Additionally, several participants discussed the positive results of workplace campaigns to increase awareness of gender disparities that supported the findings of Faulkner and Lie (2007).

Participant experiences in education and training, as well as in the workplace, identified the importance of female role models in the industry. From an educational viewpoint, the responses supported key literature elements such as the impact of positive female role models on increasing the interest of women in CS degree programs (Berg et al. 2014) or that female peers could act as role models, helping to increase retention rates (Stout et al. 2016). From a workplace viewpoint, participant responses supported literature elements regarding the lack of female representation at leadership levels led to uncertainty regarding career trajectory or the ability to meet career aspirations (Clagget, 2016; Fine, Wendt, & Carnes, 2014).

Lastly, the analysis of this category included participant reflections on what influenced their own or, potentially, another woman's decision to remain in the computing industry. Participant responses overwhelmingly supported literature elements of implicit bias (Anita Borg Institute, 2013; Buse, Hill, & Benson 2017; DuBow, 2013), identity alignment (Cundiff & Vescio, 2016; Govender & Khumalo, 2014), and self-efficacy (Beyer, 2014). Identity alignment and self-efficacy will be discussed at length in the next category. Table 13 presents potential strategies that a technology manager might implement to engage female employees in the workplace, based on study participant experiences with gender balance. Table 13 also identifies which participant(s) discussed the theme.

Table 13

Retention Strategies: Experiences with Gender Balance

Theme	Strategy
Experiences with Gender Balance	1. Create a culture of diversity (P1, P2, P6, P7, P8, P10) 2. Reinforce the need for female role models in the industry (P1, P2, P4, P10)

3. Promote self-advocacy (P9, P10)
 4. Promote the fact that women in computing jobs out-earn women in non-computing jobs (P8, P10)
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Identity alignment and self-efficacy. The most closely linked category to the theoretical framework of SIT is the identity alignment and self-efficacy category. The two themes were prevalent in both levels of the conceptual framework, acting as barriers for women seeking education and training, as well as those women seeking employment in the computing industry. These two themes also influenced women in the industry when deciding whether or not to remain in computing or leave for another opportunity.

The concept of identity alignment with perceived stereotype is closely linked with SIT's in-group/out-group dynamic. According to the literature, the further away from the perceived stereotype a woman feels, the less likely she is to pursue a career in computing (Cundiff et al., 2013). Identity alignment is also related to stereotype threat in the workplace, which leads to lower female performance due to unconscious bias (Dubow, 2013). A particularly interesting finding of this study was that while participants certainly didn't align with the gender identified by their own stereotype of a computing professional, they felt that they aligned closely with almost every other descriptive trait that they identified. Strong in-group ties of those women who gained and maintained employment in the computing industry helped to minimize identity alignment variance. This supports the findings of Faulkner and Lie (2007), who stated that women who ignored stereotypes were able to successfully gain employment in the computing industry. Additionally, participants overwhelmingly felt that they fit in with their peers.

The concept of self-efficacy is also related to the in-group/out-group dynamic of SIT. The level of difficulty women expected from educational pursuits in CS, as well as their confidence

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in their ability to be successful had a tremendous impact on their interest in computing (Govender & Khumalo, 2014). In regards to self-efficacy, participants self-reported that they felt they were of equal or greater ability than their peers. One interesting finding was that some participants did not pursue opportunities to which they felt they did not meet all of the criteria requirements, whereas their male counterparts would pursue the same opportunity even if they met very few of the requirements. Table 14 presents potential strategies that a technology manager might implement to engage female employees in the workplace, based on study participant experiences with gender balance. Table 14 also identifies which participant(s) discussed the theme.

Table 14

Retention Strategies: Identity Alignment and Self-efficacy

Theme	Strategy
Identity Alignment and Self-efficacy	1. Educate workforce about stereotyping, stereotype threat (P1, P2, P3, P4, P5, P6, P7, P8, P10) 2. Create a culture of diversity (P1, P2, P6, P7, P8, P10) 3. Reinforce the need for female role models in the industry (P1, P2, P4, P10) 4. Provide professional development opportunities (training, conferences, networking events) (P8, P9, P10)

Management and mentorship. The fourth category explored the management and mentorship experiences of the study participants. This exploration looked closely at the influence of role models on women in the industry. Role model influence was a critical element identified in the literature and this category explores that element in the context of participant experiences with managers and mentorship.

The underrepresentation of women in leadership positions was identified as a critical factor in the loss of women in the industry (Fine et al., 2014). From the perspective of SIT, positive examples of female leadership factor in heavily in the social categorization and social identification elements of in-group mindset. More specifically, having the example of a female leader in computing helps to firmly root a female employee in the in-group, as well as allowing that female employee to start modeling her behavior based on the leadership example. A strong female leadership example, in the form of a manager or mentor, can also help women in the industry to aspire to those positions, helping them to begin to align themselves with their idea of a computing leader. Additionally, managers and mentors can be critical in helping women to increase their credibility and seize opportunities by advocating for those women, providing networking connections, or even funding initiatives (Anita Borg Institute, 2013).

From a management perspective, the majority of participants had experiences with both male and female managers. Additionally, the majority of participants had no gender preference for a manager, as long as they felt that their manager was willing to support their professional goals. From a mentorship perspective, very few participants had experience with a female mentor, although more than half of the participants reported mentoring females themselves. The latter situation reinforcing the desire for female mentors, as well as highlighting the need for more senior female mentors.

Many of the participants stated that they had an interest in having a female mentor, if one was available, with several participants stating that they would like to have both male and female mentors concurrently, preferring a female mentor to seek guidance from for female specific issues, such as juggling work and motherhood. Table 15 presents potential strategies that a technology manager might implement to engage female employees in the workplace, based on

study participant experiences with management and mentorship. Table 15 also identifies which participant(s) discussed the theme.

Table 15

Retention Strategies: Management and Mentorship

Theme	Strategy
Management and Mentorship	1. Reinforce the need for female role models in the industry (P1, P2, P4, P10) 2. Make female mentors available (P1, P5, P10)

Positive and negative organizational factors. The last category explored looked at positive and negative organizational factors that had the potential to support or limit success for women in computing. Being a study that aimed on identifying strategies to help retain women in computing, identifying strategies that are already successful, at least in the minds of women who have experiences, are beneficial to present to technology managers for review. Likewise, any organizational strategies or factors that the participants viewed as limiting to their success in the industry would be equally beneficial for technology managers to be aware of, helping them to avoid employing unsuccessful strategies, or to rethink or alter current strategies that may include the identified limiting factors.

The most frequently talked about supportive organizational strategy was an employee resource group (ERG) for women. The existence of an ERG was common for participants working in a medium or large organization. Participant experiences with ERGs was overwhelmingly positive, stating that they felt that the ERGs helped to educate the entire workplace about diversity and to provide a sense of support and belonging to the women who belonged to that group. Participant experiences regarding ERGs supports the literature, which

states that ERG's created a more welcoming environment and had a positive effect on employee engagement and retention (Derven, 2014; Hui & Farnham, 2016).

Limiting factors were reported by many participants. The most common theme for these factors was related to gender isolation. Gender isolation relates to SIT in the opposite manner of identity alignment. Where identity alignment helps to confirm a sense of belonging to a particular in-group, bolstering the self-esteem linked to membership in the in-group, identity alignment variance, or a weak link to a particular in-group, can contribute to a lowered self-esteem. In many cases, participants felt the struggle not even to be recognized or rewarded, rather just to be treated the same as their male counterparts. Additionally, social behavior is often determined by in-group membership (Tajfel & Turner, 1979). One participant went so far as to change the way that she dressed because she felt that would help to increase her sense of belonging. Table 16 presents potential strategies that a technology manager might implement to engage female employees in the workplace, based on positive and negative organizational factors. Table 16 also identifies which participant(s) discussed the theme.

Table 16

Retention Strategies: Positive and Negative Organizational Factors

Theme	Strategy
Positive and Negative Organizational Factors	1. Create a culture of diversity (P1, P2, P6, P7, P8, P10)
	2. Establish an Employee Resource Group for Women (P1, P2, P4, P6)
	3. Encourage women to participate in networking groups (P1, P4, P6, P10)

Limitations of the Study

As discussed in Chapter One, this study presumed that despite the decline of women in the computing industry, women are actually interested in careers in computing. This assumption was based on the fact that prior to 1990, the number of women working in the computing industry increased at the same rate as women working in other STEM disciplines. Despite the fact that number of women in computing since 1990 has declined, the interest of women in other STEM discipline continues to increase, or at least remain stable. This indicates that there may be other factors causing the decline.

Additionally, as discussed in Chapter One, this study acknowledged several limitations. The first limitation was whether or not the participants would answer the questions honestly, without influence or personal agenda. The limitation had very little effect on the results of the study as the participants were very forthcoming with both the positive and negative aspects of their experiences in seeking a job in the computing industry. Additionally, many of the experiences shared were similar amongst participants who had no relationship beyond being a part of this study.

The second limitation was a concern that participants would not be comfortable with sharing their experiences with a member of the opposite gender. Ultimately, this limitation did not impact the study as all of the participants willingly engage in the interviews with knowledge that the person conducting the interview was male. To help ease concerns, each interview was prefaced by providing the professional and educational credentials and qualifications of the interviewer, along with the purpose of the research. Several of the participants fully supported a male researcher, citing their own belief that a solution to the identified problem required the participation of all genders.

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There were also several unforeseen limitations that emerged while conducting the research. The first unforeseen limitation was that of interview timing. Several potential participants were unable to participate due to scheduling conflicts, such as travel for the holidays. Additionally, several potential candidates were not comfortable being recorded. Due to the requirement of digital recording for transcription and analysis, these candidates were not selected. Ultimately, these emergent limitations did not impact the study because there were enough participants able to meet the required sample size and the study results did reach data saturation.

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Lastly, as discussed in Chapter One, there were two delimitations. The first delimitation was that the study focused on women currently working in the computing industry. The purpose of the study was to use the experiences of women who had successfully gained employment so that potential strategies could be developed based on their success in overcoming common obstacles.

The second delimitation was that the study was conducted only on women who worked in the computing industry in San Diego California. The study was designed to use women who had similar experiences and by focusing on women in a specific geographical area, it is more likely that they would have similar experiences. Despite the geographical focus, participants did unintentionally have diverse backgrounds, in terms of geographic representation, with four of the participants having been born and education in countries other than the U.S.

Implications for Practice

The problem that this study addressed is that technology managers lack proper strategies to retain more women in the computing industry (Beyer, 2014). This problem is concerning for several reasons. Firstly, technology managers face a shortage of qualified candidates to fill key technology positions, regardless of their gender (Kauflin, 2017; Manyika et al., 2011).

Comprising 47% of the workforce, but only 25.5% of the computing industry (Bureau of Labor Statistics, 2017) women represent a large, viable candidate pool to help fill the identified labor shortages.

Secondly, with preponderance of computing professionals being male, the computing industry is largely homogenous from a gender diversity perspective. Technology managers are missing out on the benefits of having a diverse gendered team, such as improved problem solving and creativity (DuBow, 2013) and gains from experimentation and efficiency (Gratton et al., 2007). Additionally, the lack of women representation at the team level is directly tied to the lack of women at the senior and executive levels. Women don't start out at as senior leaders and executives. With less women available to fill these roles, organizations miss out on the advantages provided by gender diverse executive boards, such as improved engagement and retention (Derven, 2014) and improved success (Hersh, 2016).

Strategies for technology managers. Table 17 presents potential strategies identified by the study that a technology manager might implement to engage female employees in the workplace. Table 17 also identifies which participant(s) discussed the theme. Discussion on how each strategy can benefit technology managers will follow.

Table

17

Retention Strategies Master List

Theme(s)	Strategy
1	1. Highlight the problem solving nature of the job (P1, P3, P4, P6, P9, P10)
1	2. Implement current or emerging technologies (P2, P7, P10)
2	3. Promote self-advocacy (P9, P10)
2	4. Promote the fact that women in computing jobs out-earn women in non-computing jobs (P8, P10)
3	

	5. Educate workforce about stereotyping, stereotype threat (P1, P2, P3, P4, P5, P6, P7, P8, P10)
3	6. Provide professional development opportunities (training, conferences, networking events) (P8, P9, P10)
4	7. Make female mentors available (P1, P5, P10)
5	8. Establish Employee Resource Group for Women (P1, P2, P4, P6)
5	9. Encourage women to participate networking groups (P1, P4, P6, P10)
2, 3, 5	10. Create a culture of diversity (P1, P2, P6, P7, P8, P10)
1, 2, 3, 4	11. Reinforce the need for female role models in the industry (P1, P2, P4, P10)

Note. Themes: 1 - Computing Background; 2 - Experiences with Gender Balance; 3 - Identity Alignment and Self-efficacy; 4 - Management and Mentorship; 5 - Positive and Negative Organizational Factors

Discussion of individual strategies will be presented using the following format:

Strategy (themes that strategy applies to). Expansion on strategy, how it applies to technology managers and workers.

Highlight the problem solving nature of the job (Computing Background). A common theme amongst study participants was their attraction to the problem solving nature of a computing job. Implementing a strategy that highlights problem solving as a workplace motivator, such as gamification, could lead to heightened engagement, in addition to improved work performance (Perrier, Celestine, Scott-ladd, & Leighton, 2016).

Implement current or emergent technologies (Computing Background). Similarly to the previous strategy, study participants were attracted to the potential to work with leading/bleeding-edge technologies. Using technology to attract and retain employees, especially those of the millennial generation, could be a valuable strategy for technology managers (Jha, Sareen, & Potnuru 2018).

Promote self-advocacy (Experiences with Gender Balance). Study participants reinforced the gender marginalization identified in the literature and often reported that they were the only female computing professional in their organization. Technology managers

(especially male) may face their own implicit bias in managing and leading female employees, putting those employees at a disadvantage (Anita Borg Institute, 2013). Implementing a strategy that promotes self-advocacy, empowering disadvantaged employees the ability to freely and comfortably speak on their own behalf, could help to identify and overcome implicit biases that exist (Daly-Cano, Meada, Vaccaro, & Newman, 2015).

Promote the fact that women in computing jobs out-earn women in non-computing jobs (Experiences with Gender Balance). There is no shortage of literature discussing the gender pay gap, which according to some sources can be as large as 25% (American Association of University Women, 2014). Several study participants noted that the salary compensation for jobs in computing was a large contributing factor towards their satisfaction with their current career choice. The participants were aware that computing jobs paid more which, according to research, could be as much as two-thirds more than women in non-computing professions (Funk & Parker, 2018a). Technology managers could help improve retention levels by (a) ensuring that their compensation levels are equal and at fair market values and (b) promoting the disparity between computing and non-computing salaries.

Educate workforce about stereotyping, stereotype threat (Identity Alignment and Self-efficacy). The existence of stereotypes, such as the belief that women were less successful with or skilled in computing proficiencies was a common occurrence reported by study participants. Stereotype threat or the decrease in the performance of female computing professionals based on their belief in the stereotype rather than their actual abilities (Hill, Corbett, & St Rose, 2010), is an issue the technology managers should be aware of and combat. Proactively educating the workforce about the danger of stereotype threats can help prevent

adverse impacts on performance, engagement, and retention for any negative stereotype, not just gender-based stereotypes.

Provide professional development opportunities (Identity Alignment and Self-efficacy). Self-efficacy, or the belief in one's own ability to succeed, was a critical barrier for women seeking computing jobs (Heilbrunner, 2013), as well as remaining in the industry (DuBow, 2013), once they found jobs. Study participants overwhelmingly responded that they felt as equal to or even more proficient than their peers, making it clear that they had a high level of self-efficacy. Those participants who did not feel as proficient were either new to the industry, new to an organization, or new to a department or job function. In those cases, participants were confident that with time, they would increase their proficiency. Providing support and funding to professional development for participants who have a low level of self-efficacy could help improve those levels over time, increasing confidence.

Make female mentors available (Management and Mentorship). Mentorship has been identified as a key factor in improving retention rate for women (Hegstad & Wentling, 2004). The majority of study participants had experience with mentorship in some form. While participants indicated that there was value in having a mentor regardless of gender, several mentioned a preference for a female mentor for gender-specific guidance. Most participants had no access to a female mentor and several participants stated that they would prefer a female mentor, if available. While an internal female mentor may not always be available, technology managers could establish a mentorship program where a female mentor from external to the organization is made available for those employees specifically looking for a female mentor.

Establish an Employee Resource Group (ERG) for Women (Positive and Negative Organizational Factors). The literature was clear regarding the positive impact that employee

resource groups (ERGs) have on employee engagement and retention (Derven, 2014; Hui & Farnham, 2016). Several of the study participants currently worked at mid-to-large sized organizations that had formal gender-based ERGs and participant support for those groups was considerable. Technology managers working at organizations that are large enough to support it, should consider a gender-based ERG. One key takeaway regarding gender-based ERGs was that participants noted that these groups were open to all genders, not just the supported gender. By being gender inclusive, the ERGs were able to bring awareness to gender-specific issues that might not be obvious to members of another gender.

Encourage women to participate in networking groups (Positive and Negative Organizational Factors). For smaller organizations that might not have enough women to support a gender-based ERG, participation in external networking groups that focus on computing specific elements could serve the same purpose as an ERG. Study participants noted that attending professional groups, even those not specifically designed for women, such as women who code or girls who code group, still afforded them the opportunity to meet and network with other women in the industry. The literature has shown that some benefits to participation in networking groups are reduced feelings of isolation (helping identity alignment), advancement of technical skills (increased, self-efficacy), and encouraged relationships (role models and potential mentors) (Anita Borg Institute, 2013). Specifically addressing the last element, some study participants noted that they were able to establish informal mentorships with other women that they found incredibly beneficial. Technology managers should consider a strategy that encourages women to participate in women's networking groups.

Create a culture of diversity (Experiences with Gender Balance, Identity Alignment and Self-efficacy, Positive and Negative Organizational Factors). Previous sections have

discussed the numerous benefits of diversity, such as increased engagement and retention (Derven, 2014), and that many Americans felt that a diverse workplace was important (Funk & Parker, 2018b). According to the literature, some organizations leverage diversity as a competitive advantage (Hunt, Dixon-Fyle, & Yee, 2018). Rather than trying to introduce diversity into an organization as a means of improvement, technology managers should create a culture of diversity that embraces differences to drive success.

Reinforce the need for female role models in the industry (Computing Background, Experiences with Gender Balance, Identity Alignment and Self-efficacy, Management and Mentorship). The last strategy to discuss emerged from analysis of several categories. So far there has been much discussion on ways to increase identity alignment and improve self-efficacy for female computing professionals. There has been discussion on the lack of female managers and mentors available and how ERGs and professional networks can help fill that need. The literature discusses the lack of role models for women who are considering computing degrees (Berg et al., 2014). Technology managers should consider implementing a strategy that reinforces the need for female role models in the industry. While the long-term benefits of influencing potential future employees does exist, there has been research that suggests that the act of mentoring could help improve the job performance and increased recognition for women who acted as mentors (Eby, Durley, Evans, & Ragins, 2006). Additionally, several study participants indicated that acting as mentors helped to tie them to the computing community and also provided as sense of pride in being able to help women overcome or avoid the same barriers that they themselves faced.

Implications of Study and Recommendations for Future Research

The findings and conclusions from this study will benefit technology managers by helping them to attract and retain qualified computing professionals, as well as realize all of the

advantages of a gender diverse team. Practical application of the 11 strategies identified or derived from the study participants should help technology managers to retain the female computing professionals that they do have, as well as create an environment that is attractive to potential qualified female candidates. Additionally, by supporting and developing the female members of their teams, it is likely that those team members will remain with the organization with the potential to transition into senior and executive leadership positions. The literature overwhelmingly links gender diverse executive boards to organizational success (Grey and Cox, 2016; Hersh, 2016; Nelson, 2014; Sonnabend et al., 2017).

The results of this study identified several areas of consideration that would benefit from further research and exploration. First, although it was not proactively part of the population and sample requirements, four of the study participants were born and received education from primary school through their undergraduate college degree in a country that was not the U.S. Specifically, they represented the countries of China, Georgia, Iran, and Italy. All four of the participants either received, or were working on graduate degrees in the U.S. Three of these study participants identified that the gender balance in the education system of their home countries was either more balanced or with a higher percentage of women than men. Further research into what factors influence the gender balance in these countries might help to identify strategies that could help improve the balance here in the U.S. Although it is criticized by some researchers, the “leaky pipeline” (Vitores & Gil-Juárez, 2016, p. 5) theory does have merit, and with more women obtaining degrees in computing and entering the workforce, there is more opportunities for a technology manager to attract and retain a qualified female computing professional. Specifically, such research could focus on identity alignment factors experienced

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by women in these countries, isolating practices, strategies, or other elements that allow these women to identify with a computing professional.

Second, several of the study participants worked for the same large organization. This organization is known for social awareness and support of diversity. Participants discussed a program in which it had existing employees reach out to candidates prior to their initial interview. During this conversation, the existing employees would share the expectations of the candidate during the interview process, as well as answer any questions the candidate might have. Female candidates were contacted by a female employee. Participants noted that this same gender contact was especially comforting, as they were able to ask gender-specific questions that they otherwise might not have asked. Additionally, these participants felt a stronger sense of loyalty their organization and department, having a sense of ownership in the team building process. Further research into this process could yield insight into the effect this practice has on retention rates. Specifically, this research could look into the effect of this gender-specific organizational strategy on the retention rates of computing professionals.

Lastly, this study focused on the experiences of women who have successfully overcome barriers and impediments in achieving their goal of obtaining a career in computing. There is much value in distilling their experiences into strategies that might be successful in both attracting women to the career field, but also to create an environment that supports engagement and retention. There have a few studies that have focused on the experiences of female engineers that have chosen to leave their career field (Singh & Fouad, 2011; Singh et al., 2013), but there seems to be no comparable literature that specifically investigates women who have left computing jobs. Comparing the experiences of women who have left the industry with those who have persisted could isolate specific strategies that might have convinced those women to stay.

Such strategies could also be used to prevent women in similar circumstances from leaving the computing industry.

The findings and conclusions from the study are part of a never ending research process focused on retaining women in the computing industry. The strategies identified a need to be further developed, tested, and deployed to understand their true level of effectiveness. This study sought to make a contribution to the literature surrounding the gender balance in the computing industry and the field of practice of technology management by exploring the experiences of women who have persisted to help create a more gender diverse and successful industry. The exploratory nature of this study allows for further research into gender balance in the computing industry.

Conclusion

The purpose of the qualitative exploratory study is to explore the strategies technology managers need to retain women in the computing industry. There are estimated shortages in crucial computing professions, such as cybersecurity, which Kauflin (2017) estimated as having a shortage of two million qualified candidates by 2019. Another growing computing discipline, big data, could face shortages of 140,000 – 190,000 workers with the appropriate skillset, according to Manyika et al. (2011). These shortages, amongst others in the industry, represent well-paying jobs that could potentially be filled by women. The problem addressed in the study was that successful strategies that technology managers need to retain women in the computing industry (Beyer, 2014). Additionally, technology managers are facing a shrinking pool of female candidates to help them create a diverse gender team. According to Corbett in Hill (2015), women comprised on 19% of computer science graduates in the United States. Even those technology managers who are proactively seeking qualified female candidates may have difficulty finding, hiring, and retaining them. This study identified 11 strategies to help increase

engagement and retention levels in female computing professionals, as well as create a more gender diverse workforce, through the analysis of qualitative data generated from 10 participant interviews conducted in San Diego, California.

The study focused on five thematic categories derived from common barriers that women seeking a career in the computing industry face. The thematic categories were: (a) computing background, (b) experiences with gender balance, (c) identity alignment and self-efficacy, (d) management and mentorship, and (e) positive and negative organizational factors. Each thematic category represented a topic that was explored in the literature review and relates to the conceptual framework developed in Chapter Two.

Participant responses were analyzed through the lens of Social Identity Theory (SIT), developed by Tajfel and Turner (1979), specifically linking the concepts of identity alignment and self-efficacy to SIT's in-group/outgroup mindset. More specifically, social identification is important because much of our behavior is patterned after our perception on how a group member should act. Additionally, self-esteem is derived from how well we feel that we match up with our idea of proper group behavior (Tajfel & Turner, 1979).

The greatest influencing factors on in-group status for women in computing discussed in the literature were identity alignment (Cundiff et al., 2013; Govender & Khumalo, 2014) and self-efficacy (Beyer, 2014). Analysis of participant responses indicated that while they did not match the stereotypical description of a computing professional that is a singularly focused, competitive, asocial, male (Lewis, Anderson, & Yasuhara, 2016), they felt they closely matched their own description computing professional, with two major caveats. All but one participant identified a stereotypical computing professional as a male, which they obviously were not. The

second deviation from the expected behavior was that most participants felt that they were more socially adept than the stereotype.

An ancillary theme in the research was that of gender diversity and the benefits that this diversity brings to an organization. From a purely descriptive statistical viewpoint, the computing industry is not gender diverse. A 2013 study showed that only 26% of computing industry workers were women (National Center for Women in Technology, 2014). In 2017, women comprised only 25.5 % of the 4.8 million computing professionals (Bureau of Labor Statistics, 2017), which indicates that the number had continued its decline. Participant experiences identified in this study supported the literature, with the preponderance of computing professionals still being male. However, some of the foreign born and educated participants did indicate that gender balance in their native countries was either equal or skewed more towards a female majority. This finding warrants more research in a future study.

This exploratory study had several limitations, including the concern that participants would answer honestly and without influence and personal agenda. Additionally, there was a concern that participants would not be comfortable sharing their experiences with a member the opposite gender. Both of these limitations did not affect the overall results of the study as all participants were comfortable in sharing their personal experiences and analysis of those experiences provided enough rich and robust data for the analysis to reach data saturation. Any dishonesty or personal agendas would have been easily identified as a deviation or outlier. An emergent limitation was the time frame for conducting data. More time would have afforded the opportunity for more participants to provide data. This limitation did not impact the results of the study because enough participants were identified to match the number of participants in similar qualitative studies. Additionally, data saturation was achieved.

The findings and conclusions from this study will benefit technology managers by helping them to attract and retain qualified computing professionals, as well as realize all of the advantages of a gender diverse team. Practical application of the 11 strategies should help improve engagement and retention levels of those female computing professional that already have, as well as attract potential qualified female candidate. The key factors in creating improvement in engagement and retention is that the identified strategies are aimed at increasing identity alignment and self-efficacy levels, firmly rooting the feeling of in-group membership. This study sought to make a contribution the literature surrounding the gender balance in the computing industry and the field of practice of technology management by exploring the experiences of women who have persisted to help create a more gender diverse and successful industry. The exploratory nature of this study allows for further research into gender balance in the computing industry.

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APPENDIX A

Informed Consent



Title of Study: Women in computing careers – Discovering the factors that attract and retain them: A qualitative research study

Investigator: Kevin Buckley

Contact Number: xxxxxxxx

Purpose of the Study

You are invited to participate in a research study. The purpose of this study is to explore the phenomenon of being a woman who has successfully obtained education and employment in the computing profession.

Participants

You are being asked to participate in the study because you have been identified as a woman who has successfully achieved higher education or training in computing, as well as successfully obtained a job in the computing industry.

Procedures

If you volunteer to participate in this study, you will be asked to do the following: Answer questions about your interest in computing education and careers, as well as share your experiences in both obtaining education and a job in computing. Additionally, you will be asked to share what barriers you faced in the above pursuits, as well as any insights into how you successfully overcame those barriers.

Benefits of Participation

There may/may not be direct benefits to you as a participant in this study. However, we hope to learn ways in which future women seeking education and careers in computing can successfully overcome any barriers that they may experience.

Risks of Participation

There are risks involved in all research studies. This study is estimated to involve minimal risk. An example of this risk is possibly feeling uncomfortable answering question about your organization.

Cost/Compensation

There will be no financial cost to you to participate in this study. The study will take approximately one hour. You will not be compensated for your time.

Contact Information

If you have any questions or concerns about the study, you may contact:

Kevin Buckley, buckleykm@gmail.com, xxxxxxxxxx – Researcher

Dr. Daniel K. Dayton, ddayton@coloradtech.edu, xxxxxxxxxx – Research Supervisor

For questions regarding the rights of research subjects, any complaints or comments regarding the manner in which the study is being conducted, you may contact Colorado Technical University – Doctoral Programs at 719-598-0200.

Voluntary Participation

Your participation in this study is voluntary. You may refuse to participate in this study or any part of this study. You may withdraw at any time without prejudice. You are encouraged to ask questions about this study at the beginning or at any time during the research study.

Confidentiality

Physical data collected will be kept in a secure, locked location when not being used or transported. Digital data will be kept on an encrypted device, protected by a multi-factor authentication. Data will be kept for seven years, after which it will be destroyed.

Participant Consent

I have read the above information and agree to participate in this study. I am at least 18 years of age. A copy of this form has been given to me.

Signature of Participant

Date

Participant Name (Please Print)

APPENDIX B

Interview Questions

Commented [u17]: What/where is question # 1?

1. Describe what led you to pursue a career in computing?
 - a. Were you encouraged to pursue this career?
 - b. What interests you most about computing?
2. Describe your background with computing training/education.
 - a. How would you describe the gender balance in your training/education?
3. In your current organization, how would you describe the gender balance?
 - a. How does this compare to previous organizations that you have worked for?
4. What is your awareness regarding the gender balance in computing?
 - a. Why do you feel that the number of women in computing is declining?
 - b. Do you feel like you will remain in the computing industry?
 - i. If yes, why will you remain?
 - ii. If no, what will cause you to leave?
5. How would you describe a typical person working in the computing industry?
 - a. How do you compare to this description?
 - b. Do you feel like you fit in with your peers in the industry?
 - c. How proficient do you feel in comparison with your peers?
6. Do you prefer to have a manager that is the same or different gender?
 - a. What factors determine this preference?
7. Describe your experiences with professional mentorship.
 - a. What experiences do you have being a mentee?

- b. What experiences do you have being a mentor?
- 8. What, if any, strategies or programs have been implemented by your current or previous managers that have contributed to your success in the computing industry?
 - a. How specifically have they helped you to succeed?
 - b. How could they be altered to further support your success?
- 9. What, if any, strategies or programs have been implemented by your current or previous managers that have limited your success in the computing industry?
 - a. How specifically have they limited your success?
 - b. How could they be altered to further support your success?
- 10. Is there anything else that you would like to add regarding your experiences with:
 - a. Computing education
 - b. Obtaining a job in computing
 - c. Remaining in the computing industry

APPENDIX C

Interview Protocol

1. Explain the purpose of the study.
2. Assure participant confidentiality and have the participant sign the informed consent agreement form.
3. Address participant physical comfort concerns (lighting, room temperature, chair, and ambient noise distraction, make water available).
4. Record the subject's number at the top of the interview field notes.
5. Encourage participants to open up about their experiences.
6. Monitor participant body language to minimize influencing subject answers.
7. Precisely record participant responses and annotate any non-verbal responses.
8. Audio record and assign a chronological number to each interview.
9. Ask interview questions in order and ask follow-on questions for clarification (see Appendix B).
10. Thank each subject for his or her participation in the study at the end of the interview.
11. Inform participants that a transcript of their interview will be made available to them when transcription is complete, and ensure participants understand they will have a final opportunity to clarify or add to responses.

APPENDIX D

Study Participant Response Practices and Strategies

The following are direct excerpts taken from the study participant responses to support each subgroup by theme. Representative practices and strategies were derived from the study participant excerpts.

Career Background: Career Interest

1. Participant P1 stated, “When I got more into coding (I code in R and Python) it was its own kind of level up to puzzle out. Now it's not only can I compute it and get the right answer but, can I make the machine do it for me? So that's kind of what appeals to me is that it's more of a game rather than like hard work.”
2. Participant P2 stated, “I think for me every design is different. And so you never get bored with it. It's always something new and so it's always a challenge every year is different because products are so different no matter where you work. The products are always different and evolving.”
3. Participant P3 stated, “Probably trying to figure out how to solve the board.”
4. Participant P4 stated, “Problem solving. I love math and problem solving ... the way the work is like you were given a problem and then you had to code something to kind of solve that. So yeah, so that was really interesting to me. So that was a really like the part that I liked. The problem solving side of things.”
5. Participant P6 stated, “I have a very logical brain and I really liked puzzles and I see a lot of the work that I do as puzzles. At the same time but when people talk to me about like what is computer science really? To me the best way to describe it is it's thinking about logic, creatively. That not only do I get to be logical but I also get to demonstrate my

creativity with the way that I solve problems. I'm not like the most artistic person in the world. So it's a good way for me to use my creative energy somewhere.”

6. Participant P7 stated, “Right now I’m focused on cloud technologies and I like how much you can do with cloud and how far you can go. Back, then I was like studying programming and I was liking programming. It varies depending on the year and what year you're asking about. If it was like back 10 years ago or like when I was a student at even earlier then cloud technologies were very new things. It was just encouraging how much you can do like in general using programing, using automation like automation of paper processes. Nowadays I'm like liking cloud and all the technologies and features the cloud gives you. How much you can develop your infrastructure and how easy you can make complicated things.”
7. Participant P9 stated, “It’s like writing something [code] and seeing if it works right away ... solving the problem.”
8. Participant P10 stated, “Problem solving. I feel that computer engineering is it is you can learn. I was usually a quick learner and I get bored quickly. I disengaged quickly and I found that this fields makes me engage all the time because all problem solving different topic, different technology every day comes out and you can jump on and learn it.”

Career Background: Career Encouragement

1. Participant P5 stated, “I would say that both my parents encouraged me to basically do whatever I wanted. My dad like to show what he did and I was the only one of five kids to take an interest in that, so he shared it with me.”
2. Participant P6 stated, “It was really important to my parents that we study STEM. My oldest sister is mechanical engineer by trade ... For me the fact that I came home and I

said, “Yeah, I want to study computer science” made my parents happy and they never bothered me about it. And you know, generally my whole family was very happy about it.”

3. Participant P7 stated, “My school teacher was thinking that I would be good in math in general and I was also encouraged by my parents. I was liking it so I can’t say that my parents were the ones who made this decision for me. I was also on the same page.”
4. Participant P8 stated, “My family is quite open-minded. They are very supportive.”
5. Participant P10 stated, “Yeah, yeah, I think differently from that time school and parents. They were definitely supportive of me.”

Career Background: Career Deterrence

1. Participant P1 stated, “I definitely was pushed more towards like, oh, you're a natural teacher that you should do that and the tech world's really hard and like competitive and I don't know if you would like flourish in that competitiveness and things like that. And yeah, so is more towards, not that I wasn't push towards math, it was more that I was being a lot more pushed towards teaching.”
2. Participant P2 stated, “I mean when I first started you were the only woman in the group you are asked to go make coffee you are asked to go make blueprints, dirty jokes abounded that were very inappropriate and you didn't have anyone to go to, you just had to stick it out, you know, and I think that's one reason there weren't many women who could stick that out and make it through.”
3. Participant P4 stated, “No one in my family really was like passionate about that. Actually I was discouraged to go to the school because it was far away from where I live there. So I had to wake up at 5:00 in the morning take three buses to get to my school.”

4. Participant P8 stated, "You're never respected but ... I get respect now but in the beginning I was never respected."

Experiences with Gender Balance: Gender Balance in Education and Training

1. Participant P1 stated, "I started getting in my stats, and they call it pure math, which is like more abstract theoretical mathematics. It would be me and this other girl Stephanie out of our entire class of 30 to 50 students."
2. Participant P2 stated, "Hmm pretty light with women. Yeah pretty light with women. I was always the only woman when I first started I started long time ago."
3. Participant P3 stated, "It's probably like two women may be out of 10. Not very many. So mostly men."
4. Participant P4 stated, "Very imbalanced. So now, it's interesting enough because when I was in Italy it was more balanced like I had a lot, like there was a good ratio between men and women in the office. Better than the ratio here."
5. Participant P5 stated, "I was usually the only girl. Sometimes there was one other girl. I remember in one the my first classes that I had in engineering school, it was for an AutoCAD class, and I was one of two girls in a hundred guys."
6. Participant P6 stated, "In high school it was bad. It was really bad. There was four women in about a class of 25 students and it was the same class. It was the same class for two years of the same exact people. We were actually the largest group of women that my teacher had had in that class because it was the same teacher who taught those all those classes for all the levels."
7. Participant P7 stated, "Back in college it was like probably 40% of women 60% of men. It was like a lot of woman were in the college."

8. Participant P8 stated, "I'm normally the only girl my entire career."
9. Participant P9 stated, "For the online trainings it's hard as I actually don't know that data. In computer science... it's been a while. I think that the majority were males. I don't remember the exact ratio."
10. Participant P10 stated, "When I went for school for my bachelor, it is very interesting the gender equality, especially for computer field is reversed back there [in Iran]. So I had more female classmates than male. So it was fun and I didn't even notice there is a gender issue in this field, but then later I went to college here and that was the reality that hit me and made me to a stop and think and make sure maybe it's not something that I want to do."

Experiences with Gender Balance: Gender Balance in the Workplace

1. Participant P1 stated, "Yeah, okay on my team there are five of us and I'm the only female. Total? Oh, wait, seven - no there's more because I forgot the UX team. [Counting] eight, nine, ten. So there's 11 women and approximately I would say rather than numbers, I would say there are, it's probably about 5 to 15% of that building as women, out of everybody there."
2. Participant P3 stated, "It's only me. I am the only female other than you know, except like admin, admin is but an email but in the design world, no just me. Out of how many people here in this office? It's probably about 10. I'm going to say about that."
3. Participant P4 stated, "Still pretty imbalanced. Probably I'd say there is 20% women."
4. Participant P6 stated, "Here we have I think we maybe have 30 women working in this office."

5. Participant P7 stated, "Previous organization would be probably a little more but not far away. So like maybe 2% or something like that."
6. Participant P8 stated, "I would say I would say there's no balance really in my personal experience. Everywhere I work. Most job sites that we're installing a job or training a customer or installing an application, I rarely see female technical staff."
7. Participant P9 stated, "My experience is definitely an imbalanced work situation. What I noticed the most is when I go to meetings when there are more managers or more senior people, you see the female presence much less than entry-level Engineers when you go to a meeting that has the more entry-level Engineers ... It's still male dominated industry. We have 35 people my organization and there are five females."
8. Participant P10 stated, "here I'm working it's actually good and that is very surprising. I can say again 6 to 4 if I want to give it a ratio, four female and six men."

Experiences with Gender Balance: Female Retention in Computing

1. Participant P1 stated, "Yes giving it[the idea] to another male coworker and I don't think that they're anybody's doing it like consciously at all. I don't know what it is. But I have been told like almost like if I do start to get like no, hey like, this is what I'm talking about. Or trying to explain myself better or like I assert myself into the conversation or the plan or whatever it is, I'll have like a co-worker gave me a jab or someone else be like be like Oh, [participant_1]'s getting riled up and but I feel like if the exact same situation, but if a man was doing that same thing it would just be oh what's John saying, you know rather than oh John's getting upset like all of a sudden it becomes I'm a bitch because I'm like trying to assert my opinion or my idea or something.."

2. Participant P1 stated, "I think it depends also on like what industry, like video games are very much male dominated. So it's very different. You almost need to be a tomboy to fit in in a sense. And so I am and so like I do fit in but you definitely like I don't see a lot of non-tomboyish types, whereas in insurance it was opposite."
3. Participant P4 stated, "I think talking with different people the idea that I get from it is it's not that women are not interested in joining this field. I think it's more of a preconception of what is expected there. Like working long hours being like, you know, because you have the image of the classic geek and so people, you know, a lot of women like sometimes they even get like into a class and they're like the only girl. And they are like, oh, I don't belong here and they change. So talking like even with girls that are now starting College. They're like, you know, like it's me and another girl and then it's all guys. Like maybe it's not the field for me like it. I feel like I don't feel like you know, I don't feel like it's a place for me. Like I don't fit there. I don't belong there and then there is the problem is that there are not a lot of Role Models. So, you know, you see Mark Zuckerberg and Bill Gates and it's always like the men. I think it's a cultural thing as well ... I feel like we as a society don't put the same effort in both."
4. Participant P4 stated, "Probably the lack of role models and support and the expectation that to be in this field. You need to be geeky."
5. Participant P5 stated, "One thing that I can think of it for my first job that I got in IT, the network engineer there flat out told that he didn't believe that women belonged in IT and that I should get out now. My first job! And that was really hard."
6. Participant P6 stated, "It can sometimes very much be this "bro" culture where it's guys are very comfortable around guys ... There are currently certain companies that I have

crossed off my list forever. Anytime a recruiter contacts me it's like absolutely not please do not contact me if you work from this company because you have a reputation and I know it. From what I hear from other people it's a toxic environment. And why do I want to work for a company where I think I'm going to be miserable.”

7. Participant P8 stated, “I would assume that the younger generation would be all into computers and gung ho on that but I’m assuming women that are not like the Millennials or whatever the heck they call them. I would assume that maybe they don't think they can. Maybe they think that's a little too much or maybe people are just technically challenged or are afraid of the computer. I know so many people that are just afraid of it for some reason.”
8. Participant P9 stated, “Based on my experience is we females tend to try to prepare a lot or thinking we aren't the perfect match for a position based on our skill set. My male co-workers might apply something if there's a couple of skills matching his skill set. What I’ve heard from my female friends is similar to my own experience, that is I look at the job description and if the majority of [the requirements] or it's not all of [the requirements] matching what I know or what I had experience with - I think that is a barrier that mindset. I have to have 100%. The skills required as has to 100% with my skill set. I think that prevented a lot of us [women] applying for a position.”
9. Participant P9 stated, “Family. That's what I can think about, families. Also maybe in some sense, after a while you kind of your career stagnates. You tend to see less females when the job levels go higher and higher and then we switch positions it when that happens and then we kind of start over new.”

10. Participant P10 stated, “I was the only female in that company and thinking that my colleagues are closed minded, not supportive. If I were in that environment, that I feel I'm the only, I get shut down. I'm not being heard and not supported. Yeah, then the differently I would consider because then I wouldn't be happy at work, that would be the reason that I would consider [leaving].”

Identity Alignment and Self-efficacy: Described Stereotype

1. Participant P1 stated, “Well, like just the stereotypes come to mind is like, they’re kind of more nerdy. Hmm, and I usually imagine like business casual attire, okay. And usually they're pretty smart, but I actually usually think like glasses. I don't know why but that usually they're pretty smart and I guess I would immediately, this sounds really bad, but I think I'd picture a Asian male immediately when you say when you use the words Computing. I know something bad about that's what most of my math cohort was Asian or white men. I feel like a lot of people that are in commute computing are pretty monotone, but really geek out on methods. So if you ask them about a specific method or technique within Computing. I think they would get very kind of excited by it. But then maybe go a little too technical into the description.”
2. Participant P2 stated, “Geeky? Typically, not very social. Detail oriented. And focused I guess. A man.”
3. Participant P3 stated, “I think a lot of us have ADD where everything has to be perfect. Okay. Well, I guess it's not a TD. It's not attention deficit disorder, but it's more like an A type. I think Type A where everything has to be a certain way we all like things done our way because there's no right way really to do it.”

4. Participant P4 stated, "That's probably white man with glasses? Probably overweight. Like it's a classic, you know representation of a geek that you even see on TV."
5. Participant P5 stated, "I would see a white dude, that's in his thirties. They tend to be not "people" people. One of the things that my dad told me was that if you can be in IT and be a "people" person, you will go really far. Especially with many of the people that I work with, they just want to create their stuff. But they're really not ["people" people] I've always been impressed with my coworkers and the amount of knowledge that they have."
6. Participant P6 stated, "I mean I see a man right. I work with mostly men. I don't know ... someone who's like really always on their computer. That stereotype of the nerd who sits in the basement and does nothing ... that nerd who sits in the dark and is on their computer. And is just like coding away and picking all the stuff up from learning from like Reddit and things like that."
7. Participant P7 stated, "Not very good at communication. They'll even maybe a little bit like introverts. They like to live in their own shell and they have that own world full of numbers rather than people. Yeah, maybe they are like they have less communicative skills because they are like, they're my well I'm talking about person who really likes what they do. So I'm thinking that they are like very focused on reading and like watching some videos and they're very focused on their profession. A man."
8. Participant P8 stated, "I would assume white male probably."
9. Participant P10 state, "I see a woman. Someone that looks like a nerd, an introvert, doesn't have friends. Someone that is not capable of talking, voicing out what they want."

They have a different way of closing and just sitting behind the computer and type, type, type, type.”

Identity Alignment and Self-Efficacy: Identity Alignment

1. Participant P1 stated, “I’m not that at all. I don’t feel like that’s what it is because I am a very loud person. I get very excited by things. I definitely do geek out and I will get if you mention a topic that I have just read about or I am working on or I remember a lot about it. I will go off and go probably way too deep into it because I get excited by it. But I definitely don’t wear glasses. I hate business casual. I’m definitely not an Asian male. So yeah, I guess that’s how I am. I think I’m pretty smart. I said that they would be really smart. But I think I’m pretty smart.”
2. Participant P2 stated, “I don’t really relate to that. I mean I’m more social but I’m in management now, so I’m not sitting doing design day-to-day anymore. I have to be more able to communicate with other groups and be a part of that where a lot of the engineers and designers they just set all day working, working, working on their on their designs. And so they don’t have to be this social because of that makes sense. That’s why I don’t fit in this industry because I’m much more social than most of them are.”
3. Participant P3 stated, “I think the industry is a lot older. So I’m probably on the younger side of the spectrum. And there are a lot of men so that kind of puts me out there, too.”
4. Participant P4 stated, “I’m the opposite and I was even told through the years, “ It’s weird like you, you know, you’re too out going to be a programmer” and I’m like what there is like now there a personality trait? This is why people sometimes like they were jokingly making you feel out of place because I like they were all “you’re too this or too that to be a programmer”. And you just like there is I don’t think there is like, you know, the job

description includes your personality traits like you can be a software engineer and look many different ways. For people that's the classic trait. Like a loner that only likes computers.”

5. Participant P5 stated, “Well, I'm a white girl. In my thirties. Pretty similar, just a different gender.”
6. Participant P6 stated, “Yeah, I don't. Do I think I'm passionate about what I do? Absolutely. Do I think I can keep up with these people most of the time? Yes. I know it's pretty symbolic of the industry to have imposter syndrome. And yes that can definitely be a factor and it can be very stressful because I don't look like the people I work with. At a certain point there are problems that I have that they are never going to have in their lives and things I have to think about that they're never going to have to think about. So I think I match up pretty well. Do I match up 100%? Absolutely not.”
7. Participant P7 stated, “Well, I'm pretty much same other than I'm a woman. I'm trying to work on my communication skills and to be nice to customers of course Sometimes I have to talk to non-IT people and explain technology to non-IT. But yeah, it's pretty much about me.”
8. Participant P8 stated, “I am very patient as you know, very patient I can get frustrated with people, but I don't show it. I'm a white female. If I'm running a job or something like that, I'm very demanding. Sometimes too much on my coworkers. I want everything to go perfect - a little OCD on that. Detail-oriented.”
9. Participant P10 stated, “I am not even close. Whatever I described actually, I'm as the opposite side of it.”

Identity Alignment and Self-Efficacy: Self-Efficacy

1. Participant P1 stated, “think I’m pretty I’m pretty even. It really depends on like, what specifically. I know where my skill set is and I know where my downfalls are. As for my particular coworkers, I would say I’m above them but they would also say that too.”
2. Participant P2 stated, “Not very anymore only because this company doesn’t do, we do more low-level kind of design work here where other companies I’ve worked at cell phone companies and other companies that are doing high tech high end things.”
3. Participant P3 stated, “I think they’ve been doing this a lot longer than I have. So I think that I may be like in the Middle where I’m still learning, but I also know a lot.”
4. Participant P4 stated, “I think like in my previous position. I felt like I was more proficient than most of the people in there. I’m usually the go to person to ask how to solve problems. And it takes time and work to prove that like I can do it and I’m probably better and you’re going to communicate with better because I’m not like the loner that doesn’t know how to talk to people.”
5. Participant P5 stated, “When it comes to Networking, for my particular job, I’m probably one of the best Network Engineers. When it comes to other subjects, I am not. I am not a SharePoint expert and they [others] can be.”
6. Participant P6 stated, “Like maybe 75 to 80%. Can I keep up with them all the time? No, but I’m also six months into this job. I still have a lot to learn.”
7. Participant P7stated, “I think that well, I don’t like to praise myself, but I’m pretty proficient. In many cases I have to transfer knowledge to my teammates who are much less proficient than I am. But of course I also have room to grow and of course there are people better than me and I learn of them from them. But overall I have pretty high rating.”

8. Participant P8 stated, "I don't feel I'm like the top dog, and I don't know if that's just laziness on my part, but I don't feel I'm the bottom either unlike kind of toward the top."
9. Participant P9 stated, "I think I'm pretty proficient but definitely early on in my career I felt like everyone was smarter than me. It takes time to get to a point or state when you figure out that people are that are good at different things and that I bring in different values. All my peers, we complement each other."
10. Participant P10 stated, "I can say that I'm really good, compared to people at the same level as me."

Experiences with Management and Mentorship: Gender Experiences and Preference

1. Participant P1 stated, "Different 100%. I don't know if ... I wonder if people would think the different same."
2. Participant P2 stated, "Doesn't matter to me. Actually, this is the first time I've ever had a woman. But it's been fine."
3. Participant P3 stated, "I've never even really thought about it. I don't know. Now that I'm thinking about it. I think I would rather have a male supervisor or boss. I guess I get along better with men than I do with women. So I think I kind of mesh a little bit better with them."
4. Participant P4 stated, "I have no preference."
5. Participant P5 stated, "I doesn't matter to me as long as I have a manager that's willing to work with me. That's all I really care about."
6. Participant P6 stated, "I have had both at this point. I've actually had a really good experience having a female manager. I think I can attribute part of that to her personality in the sense that she's incredibly empathetic and it was very, very easy to connect with

her on a personal level and also an emotional level. I find myself actually being very honest with her about the things that I need. I think a lot of this also is that I can be honest with her not just about like my work needs but also my emotional needs and then she in turn, because she's okay with me being like that.”

7. Participant P7 stated, “I don't really care to be honest. I've worked with typically with men, but I've worked with women as well and it doesn't really matter to me.”
8. Participant P8 stated, “That doesn't matter to me. You know what, I've never had a female manager though.”
9. Participant P9 stated, “I don't actually mind, but to be honest, I have not had a female manager. I work with a female manager, not directly report to her. Almost like a peer kind of relationship but from a reporting structure I have never reported to, my direct report manager. I have never had a female Manager.”
10. Participant P10, “They are the same as long as I can learn something. They are supportive, educated knowledgeable people. I don't care. But if I have a female, definitely makes me more excited to look up to that person.”

Experiences with Management and Mentorship: Mentorship Experience

1. Participant P1 stated, “I really enjoy mentoring people but I choose to Mentor women ... I guess in grad school I had a mentor that was he was a man. Then here she is not directly my, I haven't like officially asked her to be my mentor but she's like kind of my mentor. She helps me out, [she's] one of the other WITS leads.”
2. Participant P2 stated, “I've had to teach a lot of people ... I have had a lot of people over the years show me how to do things - my peers. Mostly my peers that I work with.”

3. Participant P3 stated, "I just think a lot of one-on-one interaction or teaching helped a lot and of course constructive criticism always helps too."
4. Participant P4 stated, "Kind of informally because being a director for Women who Code, a lot of people come to you and ask for questions and how to get, you know, how to move forward in their career, how to handle some types of situations ... Yeah, that's something I'm actually looking for a because the organization that I'm in now, they have a program where you can sign up and find a mentor and I'm looking for one."
5. Participant P5 stated, "I have actually taken a couple of people under my wing ... I did find [one] she wasn't in IT, but she did help me be able to grow in my career and to achieve the goals that I wanted to achieve and kind of mentor me in a more management experience. So I did get that for mentoring. One of the things that I did last year, I joined Women in IT just to kind of get a little more that community."
6. Participant P6 stated, "It was my focus more when we were running events to not so much to gain exposure to companies, but to work more on building the confidence of the woman that I went to school with to be able to be comfortable with things like interviewing and going to hackathons and you know just generally giving them a sense of community because it's something that I struggled with in University. It was something that I thought the women who were in years younger than me could benefit from having someone talk to them about ... I'm in the process of looking for a more formal mentorship. Being a formal mentee."
7. Participant P7 stated, "Basically I do the same with the Junior and middle level guys and girls. Whatever I know I teach them and like if we solve some problem together, I then tell them how we solve it why we use that approach and how that works and stuff like

that ... The senior architect who is like maybe like giving me some good tips, tricks, examples .Or he telling me something that I maybe it's easier for me to confirm him rather than talk or rather than research and like in general. We're just one team and or if I have questions I go to him and I know that he has done that before I go to him and he will teach us his what he knows about it.”

8. Participant P8 stated, “Yeah, actually, with new people, I have tried to show as much as I can on how to like work on something or if somebody doesn't know how to do something. I try to walk them through it ... I've had technicians that have mentored me.”
9. Participant P9 stated, “I have two mentees. One is a female engineer and one is a male engineer. We meet every six weeks or so mostly talking about ... they would bring in questions to me. They would receive feedback from other peers and managers and ask my opinions on [that feedback] and how they can improve [in those areas]. Mostly those kind of topics. They asked a lot about my road and how I navigated it and then improved myself ... As a mentee, I don't have formal Mentor for myself. I have a few people that I do go talk to you.”
10. Participant P10 stated, “Yes as a mentee. I have had mentors and they were men. In my previous job they were my colleagues and I've been mentoring too, but in terms of starting a study group and initiating things and sharing my goals and attracting people and sharing with them.”

Positive and Negative Organizational Factors: Supportive Strategies

1. Participant P1 stated, “I started working with them and their main goal is diversity. We work with a lot of the employee resource groups (ERGs). We have a Hispanic group, Circulo. A pan Group, which is for African-American and black employees. We have a

Pacific American [group]. We have a [Veterans group], we've got LGBTQ, etc. etc. and all of us are focused on diversity and making [the company] more diverse and more diverse culture. I really attached to WITS again because not only just in technology are women not really represented but in video games as a whole we're not represented. So I wanted to hit the ground running to like support that and what's cool is that WITS is already established and it's super supported by all the executives. So like I'm able to do these things through WITS where I'm going to women's shelter to meet and talk with some of the domestic violence victims and I get to go to the school on Friday and talk to them. And it's like my manager supports it and everybody does and they'll allow me to be passionate about women in technology.”

2. Participant P2 stated, “They actually have a group called win for women initiative Network and they have guest speakers come in and they meet maybe quarterly. I don't know how often they're always promoting to have people come to it be a part of it. Even as a manager we have to take diversity training every year as part of our goals. And that is one of the goals is the one of the diversity groups that you can choose which is what I chose. Yeah, so they're very I guess that's all I can think of off the top of my head.”
3. Participant P4 stated, “I think the mentorship program in this company is worthwhile. I haven't started, but I think it's a valuable thing. Because we have the chance to talk with people in your same road but in a senior position. So that you can move forward. Another thing is both organizations, both companies, have internal woman groups. And also really important because that you can kind of like openly talk about how you feel. It's also good because it's not like just for women. It's just a way to push inclusion in the company and I like the fact that we are able to talk about these issues”

4. Participant P5 stated, “For me, I would say, programs that have helped pay for certifications. For me, getting the certs has certainly helped me be able to master some of the field that I wanted to master. Because, some of those certs are like five hundred dollars. Being able to have that covered and being able to take that any way was really helpful.”
5. Participant P6 stated, “We have a woman's organization here called the [Company] Women Engineers organization. We have our own San Diego chapter. What's been really great about it is that it's actually put me in contact with a lot more women in this office who I don't see on a day-to-day basis and not only people who are my age but it's actually a lot more people who have been in this industry for a lot longer than I have. We had a lunch a couple weeks ago and I actually got to talk to a woman who works on the other side of office. I never see her, but she's a data scientist. I feel very comfortable being able to go to and say “do you have any career advice for me” Or I might be having this issue or I'm thinking about maybe doing this and it's very nice.”
6. Participant P7 stated, “We're trying to learn something new every two weeks. Our manager basically dedicated a few persons who would teach others. They would do the presentation of some new technology or new solution, or new vision and I will be one of them as that encourages me to deep dive because I obviously want to be very prepared for potential questions and I feel like that will educate me even better than I would educate other event because I will be the one who will be presenting. That makes me work a lot but also makes me feel that I'm getting more educated.”
7. Participant P9 stated, “We have an interview Bar Raiser Program, which would be the calibrator for interview loops and usually is not part of the hiring team. So this was, I

think, this is a very good program for someone to understand the [Company] culture through hiring processes and also build confidence by dealing with different people and different teams. My manager encouraged me to join this program. I think that over the last couple of months I have learned a lot to the through the training program and have improved the way I handle people having different opinions and then sometimes heated discussions and stuff like that.”

8. Participant P10 stated, “I remember my first job when I was an intern and they started hiring a lot more females and I felt a lot for support. They were encouraging me to go to meet ups and sometimes they would come with me ... But I feel that they see me, they wanted me to sit with them, and they were supportive of me. Which definitely helped to bring me confidence not only learn more about my field and enhance my technical skills, but at the same time feeling that they wanted me to succeed.”

Positive and Negative Organizational Factors: Limiting Factors

1. Participant P1 stated, “I guess gender isolation because it's not just about us learning and supporting each other. It's about educating both genders and supporting each other. I don't want to be viewed differently, you know, so making it just female exclusive and not male. I wanted to be invited to watch the game and go drink beer and stuff. I didn't want to go to a mani-pedi at lunch. Those kinds of things. So I definitely I don't think I like the whole “just women” exclusive because I guess that's what we're fighting for in a sense is that it's not that we want women to be necessarily recognized or listened to or whatever. It's just to be treated exactly the same. If you're not listening to the guys, that's fine. If you don't listen to me, like I get it you just kind of an asshole you don't listen to anybody. But being treated differently or being called sweetheart or being coddled or talked nicely

to just because I'm a female that's where I get more upset. I don't yeah, I don't want to be treated differently. I want them [men] to be treated exactly the same. So yeah, so I guess the whole like keeping it separate is not what I want. I want to combine as much as possible.”

2. Participant P5 stated, “The one thing that I would say that I ran into is when I did run into sexism in the workplace when I went to my bosses in HR and I was ignored.”
3. Participant P5 stated, “One of the examples I can give which was due to the company expanding beyond a certain amount. We had to do mandatory sexual harassment training. During the entire training, they singled me out for it. And were like “this is what you can't say. This is what you can't do. This is what happened to [P5].” So I sat there for the entire training basically be different than everybody else.”
4. Participant P7 stated, “Yes, in my previous company ... there was a pretty bad on-call schedule there. When I was on-call, basically they woke me up like every three hours so and it was such a crazy approach. The bad thing about it, not only not only that waking up every three hours, they were trying to have me solve problems in technology that I don't know. I can't know all technologies, right? So if there is a problem with back-up or a specific tool that I've never used, I don't know how to fix that but their approach was you're an engineer you have to research and find the problem do whatever you have to do to fix it. If you don't know the technology ... and they wake you up at 3:00 a.m. in the morning, you don't have time to learn the technology. You just do like simple things to do the break do the break/fix and then forget about it, right? So that was pretty hard and I definitely didn't succeed in learning that knowledge. I succeeded in basically finding a

quick and dirty solution, but it definitely didn't help me to learn more and added up to a lot of stress.”

5. Participant P8 stated, “They used to give most of the training to two people. The top two people and then everybody else would get nothing. What's interesting is then if one of those people left then you move up and you get most of the training and opportunities pushed at you.”
6. Participant P9 stated, “In my previous company, the conversations [with clients] - There's good and bad about this is strategy - The managers handle all the communications with business owners and some of the customers. The good thing about it was that they shielded the engineers from trauma and external destruction. The bad part about it was that it didn't give Engineers enough visibility or exposure to external stakeholders. In that case your manager would be or your only portal to the outside teams or things like that. That would limit your potential go do something beyond your immediate team and then would basically limit your ability to grow yourself into a more cross functional role.”

General Comments

1. Participant P1 stated, “I think it stems all the way down to like that the elementary kids of greeting them first is like super important. That's why I'm doing the elementary panels like it's so ingrained in our American society. I don't know about others but in in our American society, it's so ingrained that like what a woman is supposed to do, what a woman should be. We all have these underlying things that are stilled in us that we are supposed to be the caretaker of our family. Like I read this article of the age gap of first-time moms, and I'm considered an old mom and first-time mom and like that it's because I had to put off that part of my life to get a career to establish myself because if I would

have done it sooner I wouldn't have been able to do that and like a lot of those things are not fair. I understand the whole like corporate having the maternity leave not happen after a year, but like just telling me that like if I have a baby in nine months, I'm not going to be covered by maternity leave. Like that's a deterrent for me to have kids right there and I start a family and then the fact that America is so behind on like how long we get like Canada they get a freakin year Europe a lot of those countries get three years and we get 12 weeks like that's ridiculous. So it's like we already have these things are fully instilled upon us that were so young and now we're forced to make this decision of do we stay and industry or do we leave.”

2. Participant P3 stated, “You just got to keep pushing forward in this industry. That's for sure and gained a lot of confidence and what you're doing is correct and ask a lot of questions because everything is changing all the time.”
3. Participant P4 stated, “I think it's just time to think about it. I think like it's exposing even people at a younger age in exploring these options in something. Actually the woman's group of my older company was trying to do was speaking at schools. Like high school and middle school. Just to expose to the younger girls into hey, like, you know, these are the different career paths that you can have when you grow up. I think the problem is also like the way the society views thing. It's also a little bit of a how some employers or some manager can be like “this sucks now she has to go on maternity leave.” Like maybe having more of an approach that “it's fine, it's okay, you know, that's how you were born. You're a mom.” I trigger for this. So maybe having a little bit more compassion. I don't know how we can achieve that. For other women if you don't if you feel like you don't belong somewhere like because you know, your class is mainly men. So I think it's good

that I see that in most colleges there are in Girls in Computing group. Just find the support and find those role models and ask the questions. Just don't get intimidated.”

4. Participant P5 stated, “One of the things because I've been trying to recruit more women into IT and one of the things that I found is that they're resistant. They don't think that they CAN do it. They don't feel smart enough. I thought that was interesting because I heard that over and over again when talking to my friends and people.”
5. Participant P6 stated, “It's also something that we had a very honest discussion with the managers at some point about women in the workplace and how women tend to be less of a go-getter in terms of career. That they don't think about specific long-term goals, rather they're thinking more general long-term goals and it can sometimes be hurtful to career development. While we were having this conversation it was through our Woman Engineering Group. We were having this conversation with managers. One of the managers was talking about how he had written a job requirement like a job rec to go out his end. He wanted to hire more women on his team, but only men applied and it took someone else sitting down with him and reading through the requisition to say “the way that you wrote this, women aren't going to apply.” People I went to school with, and I definitely did this too, I didn't apply for a job unless I fit I thought I fit 70 to 90% of the requirements or even maybe 100% of the requirement. Sometimes that I only applied for jobs that I thought I was a fit for versus a lot of the guys I work with will apply for jobs regardless if they think they're a fit or not. They'll apply for job even if they think they fit 30% of the requirements.”
6. Participant P8 stated, “As a woman you have to have a thick skin in this industry because most men won't take you seriously unless you demand they take you seriously. Especially

in the beginning - years and years ago. I wasn't respected at all. If I was working on a job where a woman was the manager then I felt "oh my gosh! Hello. This is so refreshing to see a woman here." But if I worked with a man, they would kind of look down on me, which was interesting. If I was to say anything a woman trying to go into this industry, you have to have a thick skin. You have to be self-reliant, self-sufficient. Hopefully you don't want to be coddled. You need to demand to be taken seriously."

APPENDIX E



Dear Kevin Buckley,

Thank you for contacting AAUW about permission to use an AAUW-created figure from Solving the Equation. We're happy to provide permission to duplicate our figure "Women in selected STEM occupations, 1960-2013" in your doctoral research proposal, dissertation, and any related publications on the condition that you attribute the figure to AAUW.

You may also duplicate other figures from AAUW publications so long as your use is for academic purposes and you notify us in writing of your intent to do so.

Best wishes for the completion of your research and degree.

Sincerely,
Kevin Miller
Senior Research Associate



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From: **Mansourian, Yazdan** ymansourian@csu.edu.au

Date: Mon, Sep 17, 2018 at 9:26 PM

Subject: RE: Respectful request to use published figures

To: Kevin Buckley <buckleykm@gmail.com>

Hi Kevin,

Thank you for your email. That is alright. I am happy that you have found the paper useful. You can use all the figures published in my paper titled "exploratory nature of, and uncertainty tolerance in, qualitative research". If you require a formal letter, please provide me with some details about your school and university, then I will write that letter as well.

All the best

Yazdan

From: Kevin Buckley [mailto:buckleykm@gmail.com]

Sent: Tuesday, 18 September 2018 3:33 AM

To: Mansourian, Yazdan <ymansourian@csu.edu.au>

Subject: Respectful request to use published figures

Dr. Mansourian,

I am a doctoral student currently working on my dissertation on gender balance in computing. I would very much like to use several of the figures published in your paper titled " Exploratory nature of, and uncertainty tolerance in, qualitative research". As my dissertation will ultimately be published, I am respectfully requesting permission to use and reprint these figures, with proper citation and reference to your published work.

If you do not have an issue with this usage, would it be possible to obtain a formal authorization letter, preferably in electronic format, to include in my dissertation documentation? This would be greatly appreciated.

Warmest Regards,
Kevin Buckley